



Asociación Colombiana de
Neumología y Cirugía de Tórax



VI SIMPOSIO REGIONAL ACMI

— SAN GIL — SANTANDER —

Homenaje al Dr. Franklin Quiroz





 **26 y 27**
de junio de 2026

 Hotel Mundo Guarigua
San Gil



REGISTRO E
INSCRIPCIONES

   ACMI_SANTANDER











"Educación continua que transforma la práctica clínica"

Capítulo ACMI Santander

VI Simposio Regional ACMI

San Gil • Santander • 26 y 27 junio 2026
Hotel Mundo Guarigua

Homenaje al Dr. Franklin Quiroz





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VIERNES 26 DE JUNIO DE 2026

08:00 - 08:30 Registro e inscripciones

08:30 - 09:00 Inauguración
Dr. Alexander Jerez
Presidente ACMI Santander

 **BLOQUE I Talleres prácticos | 08:30 - 12:00**

- Insulinización
Dr. Julio Oliveros
Especialista en Medicina Interna
- Gases arteriales
- Espirometría

 **12:00 - 14:00** Almuerzo

CONFLICTO DE INTERES

NO ES UNA CHARLA PATROCINADA / NO EXISTEN CONFLICTOS DE INTERES



IDF DIABETES ATLAS 2025

11.ª edición – Resumen de estadísticas clave

El informe más completo sobre el impacto de la diabetes en el mundo.



589
MILLONES

de adultos (20-79 años)
viven con diabetes en
el mundo (1 de cada 9)



3,4
MILLONES

de muertes al año
atribuidas a la diabetes



1 BILLÓN
DE USD

en gasto sanitario por
diabetes en adultos



252
MILLONES

de adultos con diabetes
no están diagnosticados



4
DE CADA 5

personas con diabetes
viven en países de
ingresos bajos y medios

Proyección 2050:
853 MILLONES
(1 de cada 8)



1 muerte
cada 9 segundos



Aumento del
338% en las
últimas décadas



1 de cada 2
personas con diabetes
no lo sabe



La carga económica
sigue aumentando
globalmente

ADULTOS (20-79 AÑOS) CON DIABETES POR REGIÓN – 2025



	Personas con diabetes (millones)	Proyección 2050 (millones)
Pacífico Occidental	215	260
Sudeste Asiático	107	152
Oriente Medio y Norte de África (MENA)	85	136
Europa	64	67
América del Norte y Caribe (NAC)	63	76
América del Sur y Central (SACA)	33	49
África	24	55
TOTAL MUNDIAL	589	853

TENDENCIAS CLAVE



La diabetes sigue en aumento en todo el mundo.



El crecimiento más rápido se observa en países de ingresos medios y bajos.



Una gran proporción de personas sigue sin diagnóstico.



El gasto en salud por diabetes sigue siendo muy alto y va en aumento.

TIPOS DE DIABETES

90-95%

Diabetes tipo 2



5-10%

Diabetes tipo 1

Otros tipos específicos

IMPACTO GLOBAL



Mayor riesgo de complicaciones cardiovasculares, renales, oculares y amputaciones.



Afecta la calidad de vida y la productividad de las personas y sus familias.



Requiere acción urgente, políticas públicas efectivas y acceso equitativo a la atención.



MENSAJE CLAVE

Actuar ahora para prevenir, diagnosticar a tiempo y garantizar el acceso a la atención puede cambiar el futuro de la diabetes.

CUANDO HABLAMOS DEL USO DE INSULINA COMO TRATAMIENTO DE LA DIABETES



MATT GROENING

ART 5490-15



Article: Care Delivery

Insulin adherence behaviours and barriers in the multinational Global Attitudes of Patients and Physicians in Insulin Therapy study

M. Peyrot^{1,2}, A. H. Barnett³, L. F. Meneghini⁴ and P.-M. Schumm-Draeger⁵

¹Department of Sociology, Loyola University of Maryland; ²Department of Medicine, Johns Hopkins University, Baltimore, MD, USA; ³University of Birmingham and Biological Research Unit, West of England NHS Foundation Trust, Birmingham, UK; ⁴Department of Medicine, University of Miami, Miami, FL, USA and ⁵Department of Medicine, Bogenhausen Academic Teaching Hospital, Munich, Germany

Accepted 11 January 2012

Diabet. Med. 29, 682–689 (2012)



Hypoglycemia and Risk of Cardiovascular Disease and All-Cause Mortality in Insulin-Treated People With Type 1 and Type 2 Diabetes: A Cohort Study

Diabetes Care 2015;38:316–322

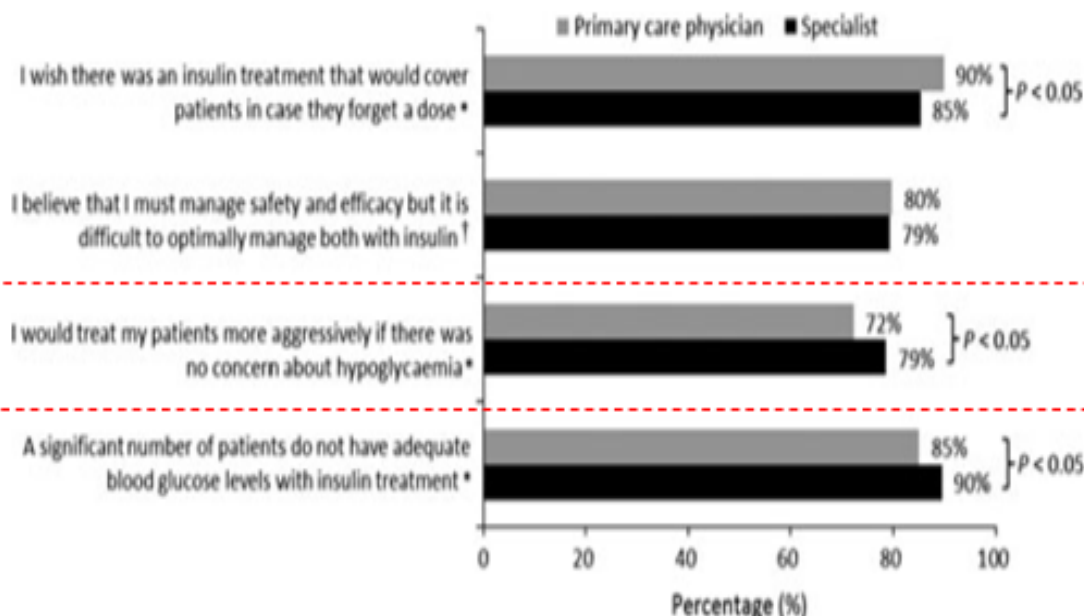
Kamlesh Khunti,^{1,2} Melanie Davies,^{1,2}
Azem Mujed,³ Brian Larsen Thorsted,⁴
Michael Lyng Wolden,⁴ and Sanjoyk Paul⁵



Table 3—Unadjusted incidence rates and 95% CIs of CV outcomes and all-cause mortality

Population			Unadjusted incidence rates (per 1,000 person-years)	Unadjusted HR (CPRD plus HES)
Type 2 diabetes	CV events	History of CVD before index (n = 1,261, events = 232)	45.3 (39.5, 51.2)	1.80* (1.17, 2.77)
		No CVD before index (n = 9,161, events = 740)	16.2 (15.1, 17.4)	1.73* (1.38, 2.18)
	All-cause mortality	History of CVD before index (n = 1,261, deaths = 510)	90.0 (82.2, 97.8)	2.58* (2.04, 3.27)
		No CVD before index (n = 9,161, deaths = 1,884)	39.7 (37.9, 41.5)	3.24* (2.90, 3.63)

Unadjusted and adjusted HRs and 95% CIs for composite CV events and all-cause mortality in patients with type 1 and type 2 diabetes, separately for those with and without a history of CVD prior to the index date. *P < 0.001. †P < 0.05.



¿PORQUE INSULINIZAR?



-1%
EN HbA_{1c}
PUEDA CAMBIAR EL PRONÓSTICO



21%
Muertes por Diabetes



14%
Infarto agudo
del Miocardio

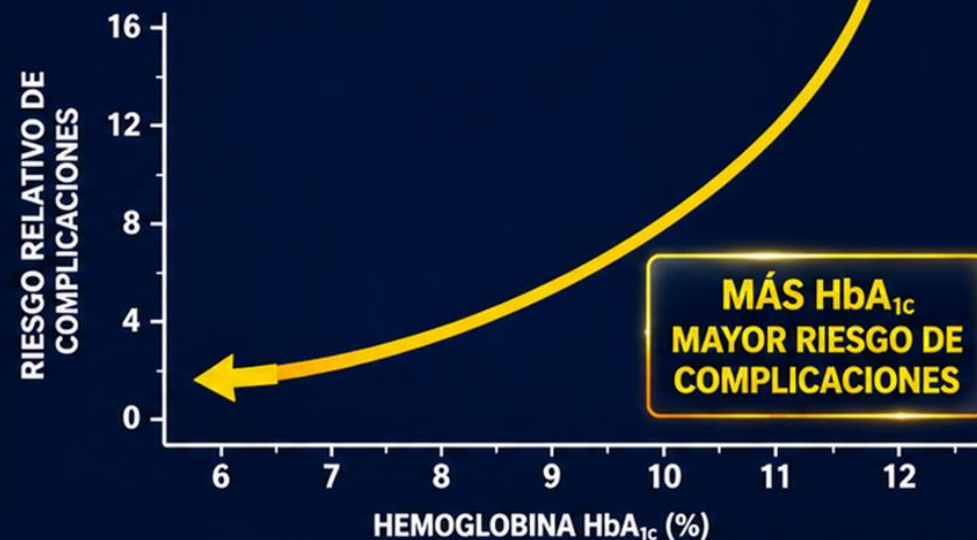


37%
Complicaciones
microvasculares



43%
Enfermedad vascular
periférica

¿POR QUÉ INSULINIZAR?



Glucosa Promedio mg/dl	120	150	180	210	240	270	300
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MEJOR CONTROL = MEJORES RESULTADOS



**MENOS EVENTOS
CARDIOVASCULARES**



**MENOS
COMPLICACIONES**



**MAYOR CALIDAD
DE VIDA**



The Legacy Effect in Type 2 Diabetes: Impact of Early Glycemic Control on Future Complications (The Diabetes & Aging Study)

Diabetes Care 2019;42:416–426 | <https://doi.org/10.2337/dci.17.1.414>

Neda Laikempong,¹ Sandra A. Ham,²
Yue Gao,¹ Howard H. Maffei,³
Jennifer Y. Wu,⁴ Elbert S. Huang,¹ and
Andrew J. Karter⁵

OBJECTIVE

To examine for a legacy effect of early glycemic control on diabetic complications and death.

RESEARCH DESIGN AND METHODS

This cohort study of managed care patients with newly diagnosed type 2 diabetes and 10 years of survival (1997–2013, average follow-up 13.0 years, $N = 34,737$) examined associations between $HbA_{1c} < 6.5\%$ (< 48 mmol/mol), 6.5% to $< 7.0\%$ (48 to < 53 mmol/mol), 7.0% to $< 8.0\%$ (53 to < 64 mmol/mol), 8.0% to $< 9.0\%$ (64 to < 75 mmol/mol), or $\geq 9.0\%$ (≥ 75 mmol/mol) for various periods of early exposure (0–1, 0–2, 0–3, 0–4, 0–5, 0–6, and 0–7 years) and incident future microvascular (end-stage renal disease, advanced eye disease, amputation) and macrovascular (stroke, heart disease/failure, vascular disease) events and death, adjusting for demographics, risk factors, comorbidities, and later HbA_{1c} .

RESULTS

Compared with $HbA_{1c} < 6.5\%$ (< 48 mmol/mol) for the 0-to-1-year early exposure period, HbA_{1c} levels $\geq 6.5\%$ (≥ 48 mmol/mol) were associated with increased microvascular and macrovascular events (e.g., HbA_{1c} 6.5% to $< 7.0\%$ [48 to < 53 mmol/mol] microvascular hazard ratio 1.204 [95% CI 1.063–1.365]), and HbA_{1c} levels $\geq 7.0\%$ (≥ 53 mmol/mol) were associated with increased mortality (e.g., HbA_{1c} 7.0% to $< 8.0\%$ [53 to < 64 mmol/mol]: 1.290 [1.104–1.507]). Longer periods of exposure to HbA_{1c} levels $\geq 8.0\%$ (≥ 64 mmol/mol) were associated with increasing microvascular event and mortality risk.

CONCLUSIONS

Among patients with newly diagnosed diabetes and 10 years of survival, HbA_{1c} levels $\geq 6.5\%$ (≥ 48 mmol/mol) for the 1st year after diagnosis were associated with worse outcomes. Immediate, intensive treatment for newly diagnosed patients may be necessary to avoid ineliminable long-term risk for diabetic complications and mortality.

In the U.S., an estimated 1.4 million adults are newly diagnosed with diabetes every year and present an important intervention opportunity for health care systems. In patients newly diagnosed with type 2 diabetes, the benefits of maintaining an $HbA_{1c} < 7.0\%$ (< 53 mmol/mol) are well established. The UK Prospective Diabetes Study (UKPDS) found that a mean HbA_{1c} of 7.0% (53 mmol/mol) lowers the risk of

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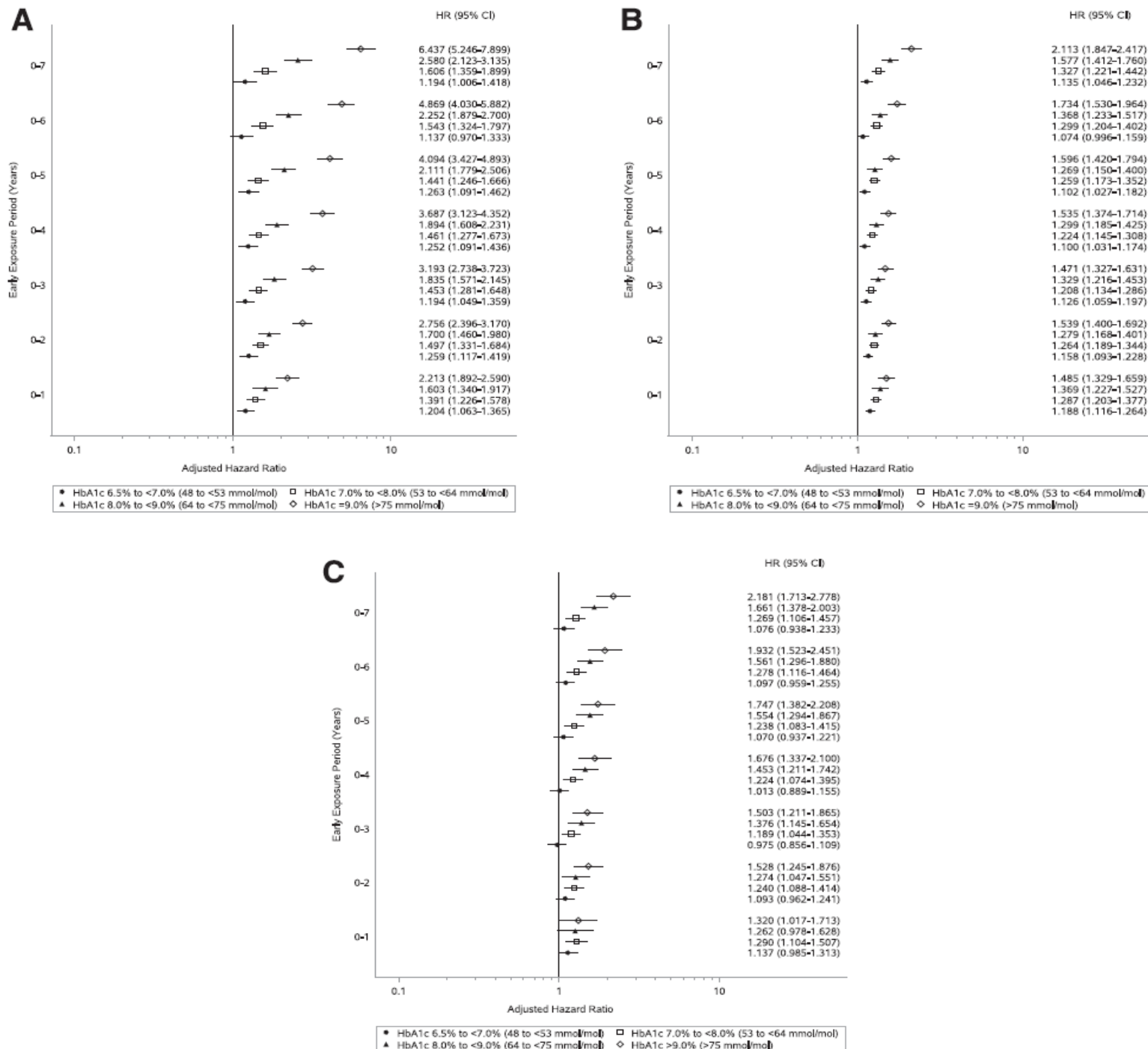
Received 8 June 2018 and accepted 7 July 2018

This article contains Supplementary Data online at <http://care.diabetesjournals.org/lookup/suppl/doi:10.2337/dci.17.1.414/-/DC1>.

This article is featured in a podcast available at <http://www.diabetesjournals.org/content/diabetes-care-updates-podcasts>.

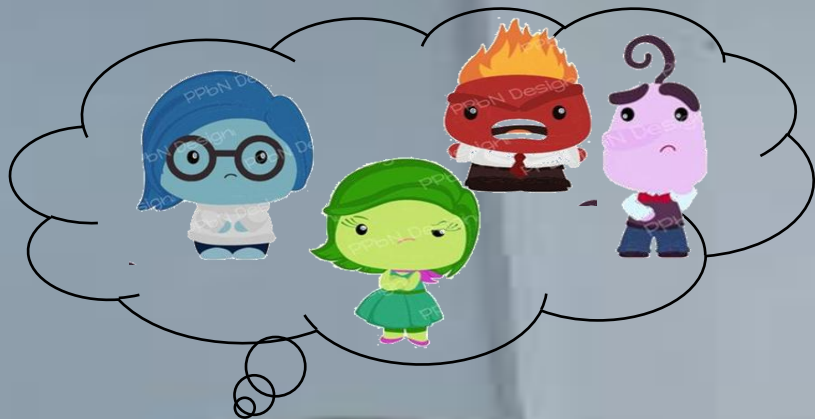
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See accompanying article, p. 348.



Diabetes Care 2019;42:416–426

Figure 2—A: Microvascular events (vs. HbA_{1c} 6.5% [48 mmol/mol]). B: Macrovascular events (vs. HbA_{1c} 6.5% [48 mmol/mol]). C: Mortality (vs. HbA_{1c} 6.5% [48 mmol/mol]). HRs adjusted for year of diagnosis, age at diagnosis, sex, race/ethnicity, BMI, systolic and diastolic blood pressure, total cholesterol, HDL cholesterol, smoking status, HbA_{1c} after each early exposure period, and comorbidity



CONSIDERACIONES PARA EL USO DE INSULINA

Criterios que orientan el inicio de insulino terapia



VALORES DE LABORATORIO

- **HbA1c** al diagnóstico >10% (ADA) o >9% (AACE)
- **Glucosa en sangre** >300 mg/dl
- **HBA1c 2%** por encima de la meta individualizada, a pesar de dosis máximas de ADOs



CRITERIOS CLÍNICOS

- Uso prolongado de corticosteroides
- Síntomas de hiperglicemia o evidencia de catabolismo
- Intolerancia a antidiabéticos orales



DIAGNÓSTICO Y HOSPITALIZACIÓN

- Hospitalización
- Si el DX es DMT1, es la única posibilidad de tratamiento
- Diabetes gestacional



COMPLICACIONES ASOCIADAS

- Insuficiencia hepática
- Insuficiencia renal avanzada
- Complicaciones agudas de la diabetes
- Infecciones y enfermedades intercurrentes

TORONTO
10 C.C. VIAL
per C.C.

JAN 21 '25

LABORATORIES
TORONTO, TORONTO, CAN.

INSULIN, TORONTO
10 C.C. VIAL
Units per C.C.

JAN 21

292-19

CONNAUGHT LABORATORIES
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INSULIN, TORONTO
100 UNITS—10 C.C.
10 Units per C.C.

guaranteed until

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10 Units per C.C.

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INSULINAS ACTUALES Y EMERGENTES

INSULINAS PREPRANDIALES

INSULINAS DE
ACCION
ULTRA LENTA

INSULINA LISPRO
U 100

INSULINA GLULISINA
U 100

INSULINA ASPART
U 100

INSULINAS
HUMANAS
ACCION RAPIDA

INSULINA
CRISTALINA
U100

INSULINAS HUMANAS
ACCION INTERMEDIA

INSULINA NPH
U -100

INSULINAS BASALES

ANALOGOS DE
INSULINA
ACCION LENTA

INSULINA GLARGINA
U- 100

INSULINA DETEMIR
U-100

BIOSIMILAR
INSULINA GLARGINA
U100

ANALOGOS DE INSULINA
ACCION
ULTRA LENTA

INSULINA
GLARGINA U300

INSULINA
DEGLUDEC U 100

INSULINA
DEGLUDEC U 200

INSULINA ICODEC
U100

INSULINAS COMBINADAS

INSULINAS
PREMEZCLADAS

INSULINA ASPART
MIX 70/30

INSULINA LISPRO
MIX 50/50

INSULINA LISPRO
MIX 75/25

INSULINA + GLP 1

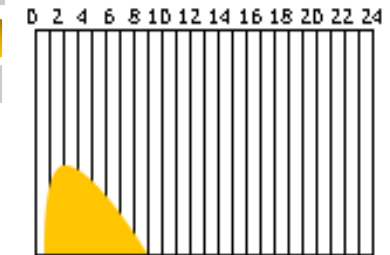
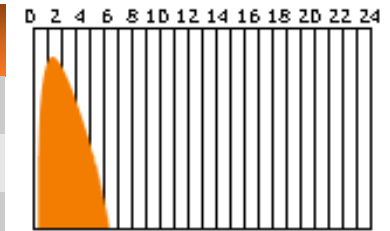
INSULINA DEGLUDEC 100
UI ML+LIRAGLUTIDE 3,6
MG/ML

INSULINA GLARGINA 100
UI/ML + LIXISENATIDE
33ug/ML 30-60

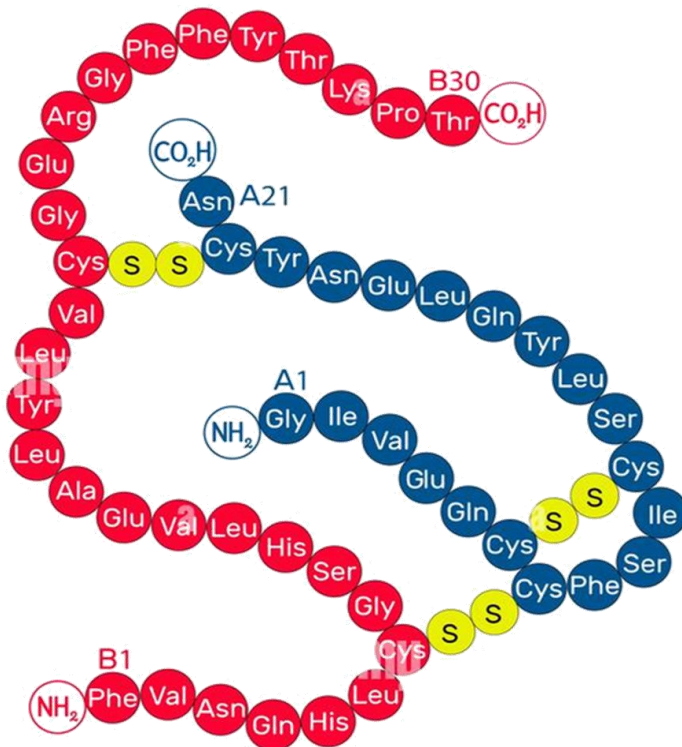
INSULINA GLARGINA
100UI/ML + LIXISENATIDE
50ug/ML 10-40

PERFILES FARMACOCINÉTICOS DE LAS INSULINAS

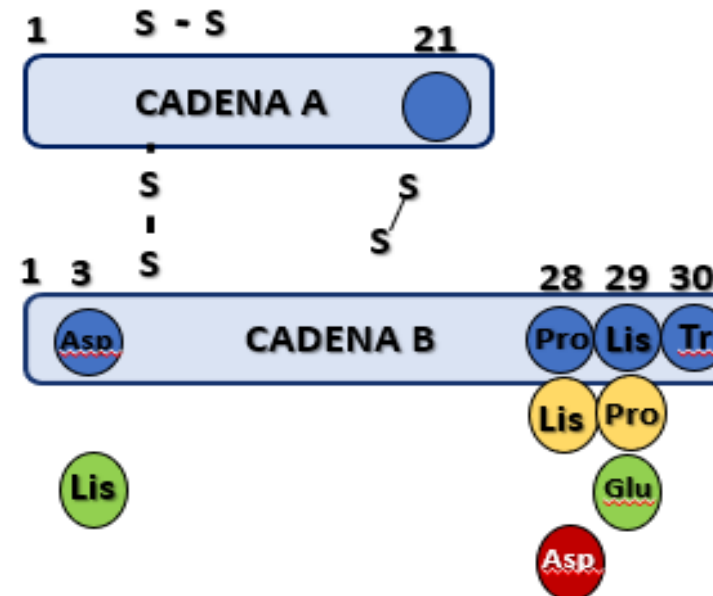
INSULINAS ULTRARAPIDAS	INICIO	PICO MAXIMO	FINAL
INSULINA LISPRO	5-15MINUTOS	1-2 HORAS	4 HORAS
INSULINA GLULISINA	5-15MINUTOS	1-2 HORAS	4 HORAS
INSULINA ASPART	5-15MINUTOS	1-2 HORAS	4 HORAS
INSULINA RAPIDA	INICIO	PICO MAXIMO	FINAL
REGULAR O CRISTALINA	30- 60 MINUTOS	2-4 HORAS	6-8 HORAS



ESTRUCTURA PRIMARIA DE LA INSULINA HUMANA



ANALOGOS DE INSULINA. MODIFICACIONES DE LA INSULINA NATURAL PARA ALTERAR SU PERFIL FARMACOCINETICO



Gli – Glicina
 Glu – Ac.glutamico
 Asn - Aspargina
 Asp- Ac. Aspartico
 Arg - Arginina
 Pro - Prolina
 Lis - Lisina
 Tr- treonina

INSULINA

- Insulina Lispro
- Insulina Glulisina
- Insulina Aspartica



Management of Hyperglycemia in Type 2 Diabetes, 2022.

A Consensus Report by the American Diabetes Association (ADA) and the European Association for the Study of Diabetes (EASD)

<https://doi.org/10.2337/doi22-0034>

Melanie J. Davies,^{1,2} Varsha R. Aroda,³
Billy S. Collins,⁴ Robert A. Gabbay,⁵
Jennifer Green,⁶ Nisa M. Maruthur,⁷
Sylvia E. Rosas,⁸ Stefano Del Prato,⁹
Charal Mathieu,¹⁰
Giuseppe Mingrone,^{11,12,13}
Peter Rossing,^{14,15} Tsvetelina Tankova,¹⁶
Apostolos Tsapas,^{17,18} and John B. Buse¹⁹



The American Diabetes Association and the European Association for the Study of Diabetes convened a panel to update the previous consensus statements on the management of hyperglycemia in type 2 diabetes in adults, published since 2006 and last updated in 2019. The target audience is the full spectrum of the professional health care team providing diabetes care in the U.S. and Europe. A systematic examination of publications since 2018 informed new recommendations. These include additional focus on social determinants of health, the health care system, and physical activity behaviors, including sleep. There is a greater emphasis on weight management as part of the holistic approach to diabetes management. The results of cardiovascular and kidney outcomes trials involving sodium-glucose cotransporter 2 inhibitors and glucagon-like peptide 1 receptor agonists, including assessment of subgroups, inform broader recommendations for cardiorenal protection in people with diabetes at high risk of cardiorenal disease. After a summary listing of consensus recommendations, practical tips for implementation are provided.

Type 2 diabetes is a chronic complex disease, and management requires multifactorial behavioral and pharmacological treatments to prevent or delay complications and maintain quality of life (Fig. 1). This includes management of blood glucose levels, weight, cardiovascular risk factors, comorbidities, and complications. This necessitates that care be delivered in an organized and structured way, such as described in the chronic care model, and includes a person-centered approach to enhance engagement in self-care activities (1). Careful consideration of social determinants of health and the preferences of people living with diabetes must inform individualization of treatment goals and strategies (2).

This consensus report addresses the approaches to management of blood glucose levels in nonpregnant adults with type 2 diabetes. The principles and approach for achieving this are summarized in Fig. 1. These recommendations are not generally applicable to individuals with diabetes due to other causes, for example, monogenic diabetes, secondary diabetes, and type 1 diabetes, or to children.

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⁵American Diabetes Association, Arlington, VA

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¹¹Università Cattolica del Sacro Cuore, Rome, Italy

¹²Fondazione Policlinica Universitaria A. Gemelli (IRCCS), Rome, Italy

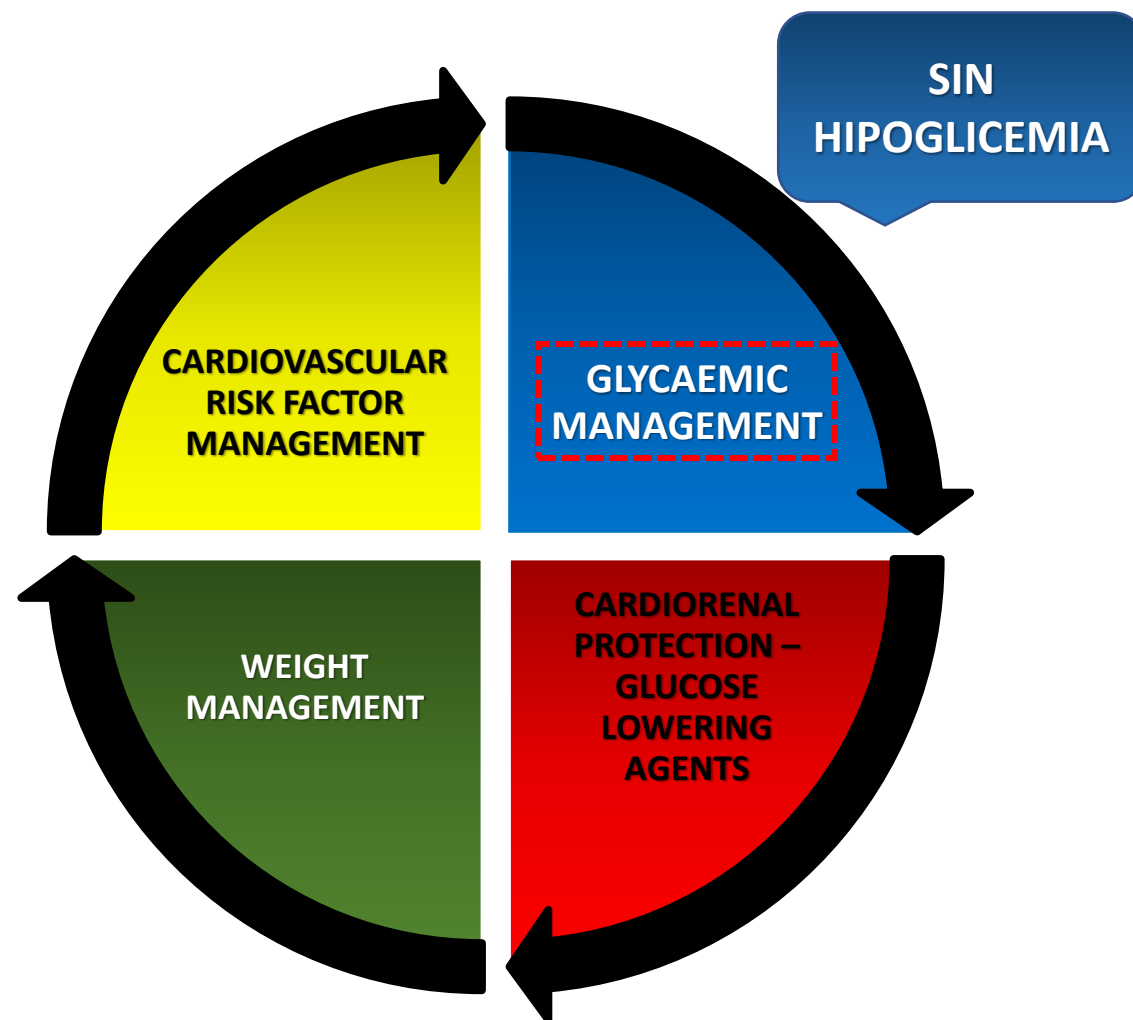
¹³Division of Diabetes and Nutrition of Scientific School of Cardiovascular and Metabolic Medicine and Sciences, King's College London, London, U.K.

¹⁴Steno Diabetes Center Copenhagen, Herlev, Denmark

¹⁵Department of Clinical Medicine, University of Copenhagen, Copenhagen, Denmark

¹⁶Department of Endocrinology, Medical University—Sofia, Sofia, Bulgaria

MANEJO DE LA HIPERGLICEMIA Y PREVENCIÓN DE COMPLICACIONES



Diabetes Care.

JANUARY 2026 | VOLUME 49 | SUPPLEMENT 1
DIABETESJOURNALS.ORG/CARE



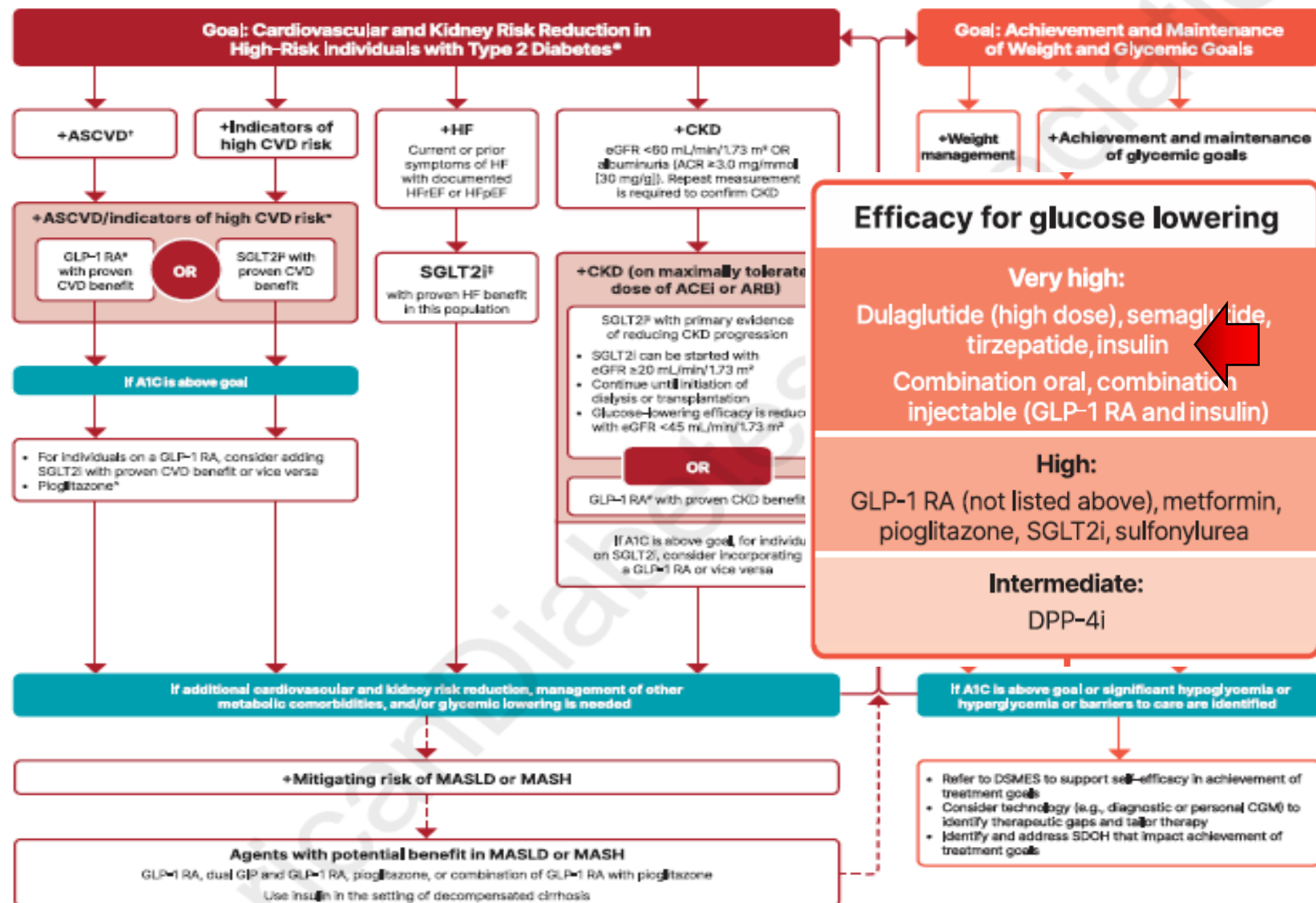
Standards of Care in Diabetes—2026



Use of Glucose-Lowering Medications in the Management of Type 2 Diabetes

HEALTHY LIFESTYLE BEHAVIORS; DIABETES SELF-MANAGEMENT
EDUCATION AND SUPPORT; SOCIAL DETERMINANTS OF HEALTH

To avoid therapeutic inertia, reassess and modify treatment regularly (3–6 months)



Diabetes Care

JANUARY 2026 | VOLUME 49 | SUPPLEMENT 1

DIABETESJOURNALS.ORG/CARE



VOLUME 49 | SUPPLEMENT 1 | PAGES S1-S371

Standards of Care in Diabetes—2026

American Diabetes Association

ISSN 0149-5992

Medication (route of administration)	Glucose-lowering efficacy ¹	Hypoglycemia risk	Weight effects ²	CV effects		Kidney effects		MASH effects	Clinical considerations and adverse effects
				Effect on MACE	Effect on HF	Progression of CKD	Dosing/use considerations*		
Insulin (human) (SQ; regular insulin also available as inhaled formulation)	High to very high	Yes	Gain	Neutral	Neutral	Neutral	Lower insulin doses required with a decrease in eGFR; titrate per clinical response	Unknown	- Injection site reactions - Higher risk of hypoglycemia with human insulin (NPH or premixed formulations) vs. analogs - Risk of hypoglycemia and duration of activity increases with the severity of impaired kidney function. - Refer to device-specific instructions for insulins compatible with different delivery systems (i.e., pumps, connected insulin pens, insulin patches).
Insulin (analogues) (SQ)									

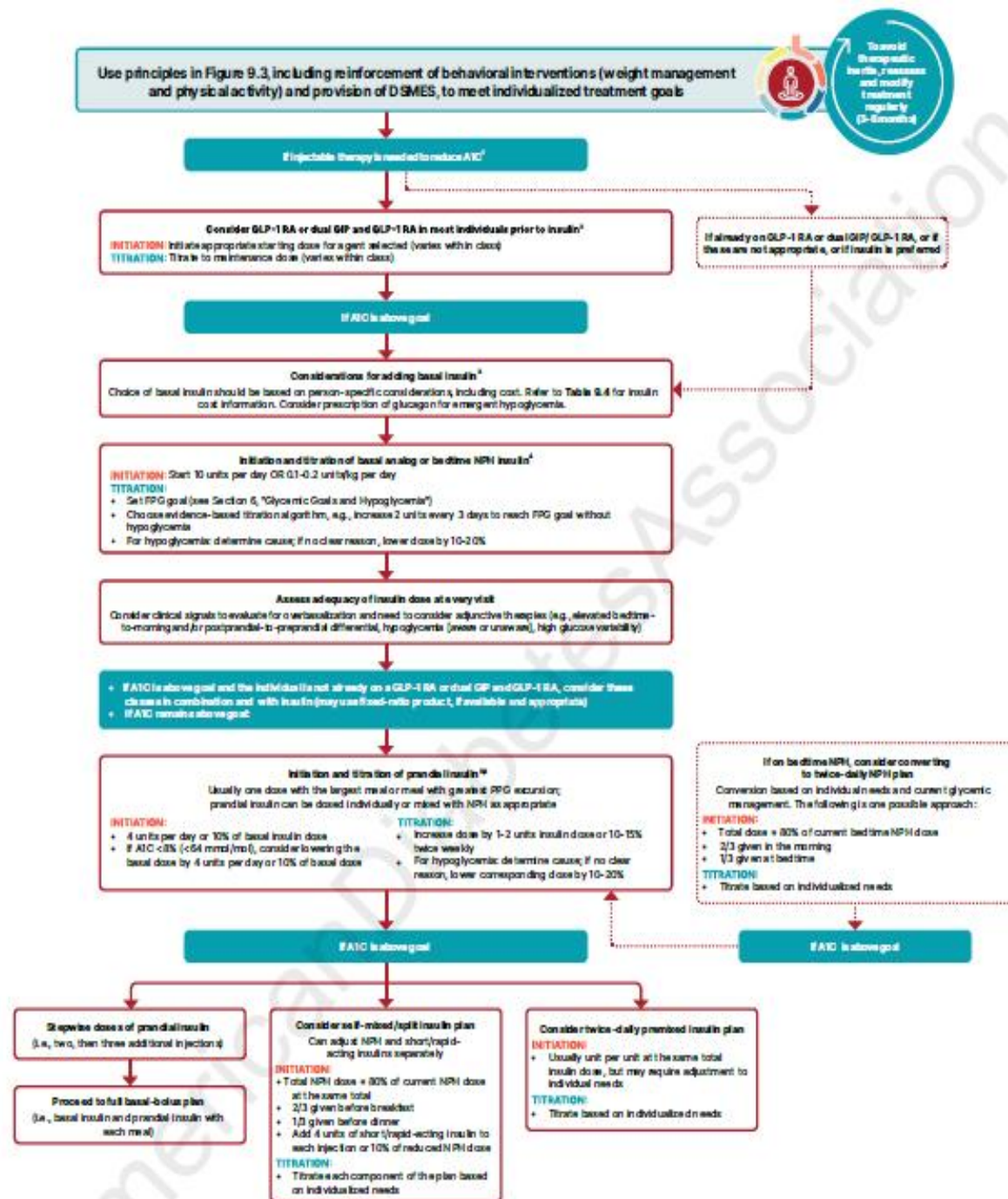
Diabetes Care.

JANUARY 2026 | VOLUME 49 | SUPPLEMENT 1
DIABETESJOURNALS.ORG/CARE



Standards of Care in Diabetes—2026

American Diabetes Association.
ISSN 0149-5992





CASO 1 : INTENSIFICACIÓN EN PERSONAS NO CONTROLADAS ADECUADAMENTE (HBA 1c > 7%) CON ANTIDIABETICOS ORALES

Perfil clínico

Nombre : **CELSO GORGORY** Edad: 66 años

APP Diabetes Mellitus T2, HTA, Dislipidemia.

Duración de la diabetes : 18 años

Talla: 1,68 m Peso: 105 Kg IMC: 37,2 Kg / m₂

Resultados de Laboratorio:

HbA1c 9,6 %; Creatinina 1,35 mg/dL R- A/C 36 mg/g ; Glucosa 168 mg/ dL
Post prandial 306 mg/dL; c-LDL 68 mg/dL eTFG x CKD: 54.3 ml/ min/ 1,73m²

Medicación actual: Empagliflozina + Metformina 12.5/1000mg 2v día,
Semaglutida 14mg día, Valsartan + Clortalidona 80/12.5 mg,
Rosuvastatina 20mg día.





Sr. GORGORY

HbA1c 9,6 %

POST PRANDIAL 306 mg/dL

Categoría de riesgo **MUY ALTO**

IMC: **37,2** Kg / m²

eTFG E IIIA / V A2

3 ADOs Dosis Máxima

Necesita terapia inyectable para reducir A1C

Considere GLP-1 RA o GIP/GLP-1 RA en la mayoría de las personas antes de la insulina.

INICIACIÓN: Iniciar la dosis inicial adecuada para el agente seleccionado (varía dentro de la clase)

TITULACIÓN: Titular hasta la dosis de mantenimiento (varía dentro de la clase)

Si está por encima del objetivo de A1C

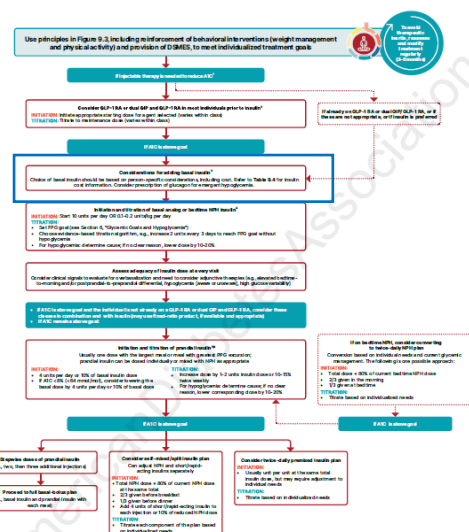
Si ya está en GLP-1 RA o GIP dual y GLP-1 RA o si estos no son Apropriado. Se prefiere la insulina.

ANADIR INSULINA BASAL

La elección de la insulina basal debe basarse en consideraciones específicas de la persona, incluido el costo. Considere la prescripción de glucagón para Hipoglucemia emergente.

¿POR QUÉ?

INSULINA BASAL



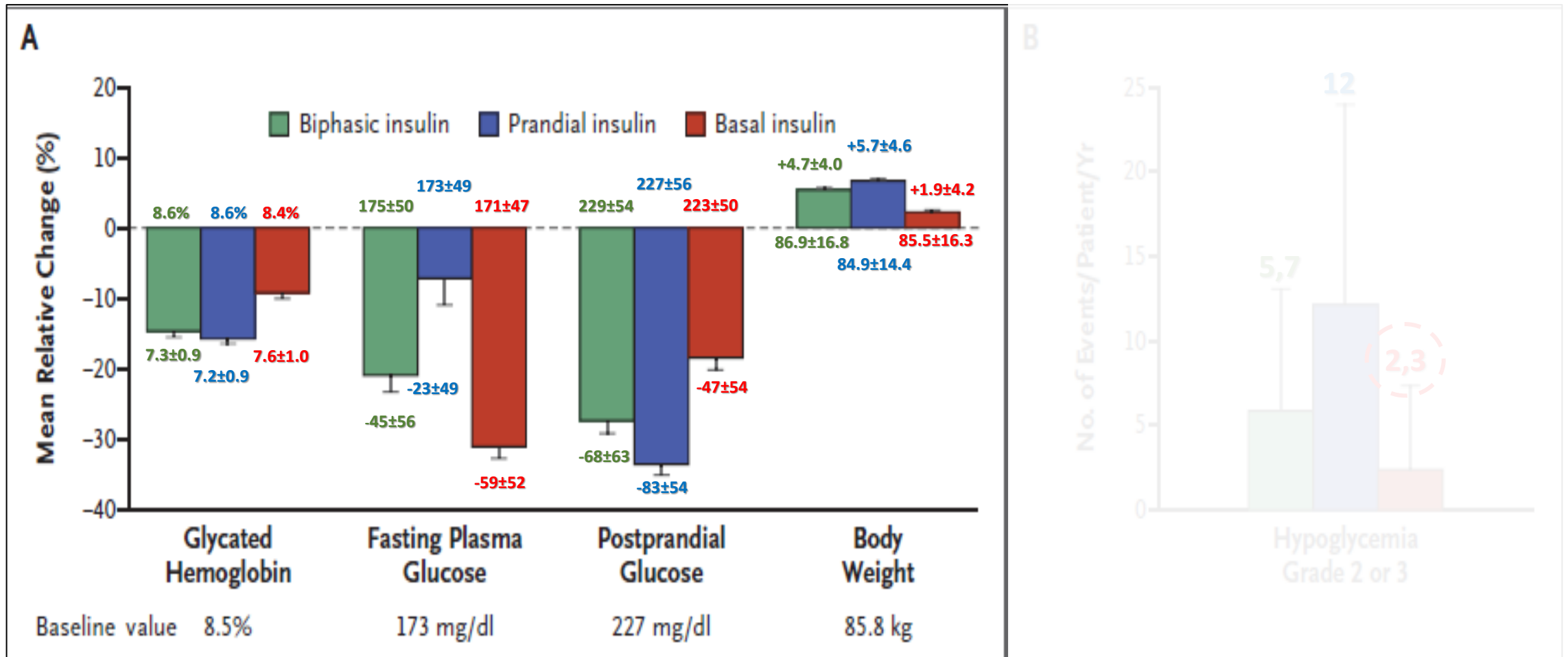
ORIGINAL ARTICLE

Addition of Biphasic, Prandial, or Basal Insulin to Oral Therapy in Type 2 Diabetes

Rury R. Holman, M.B., Ch.B., F.R.C.P., Kerensa I. Thorne, M.Sc.,
Andrew J. Farmer, D.M., F.R.C.G.P., Melanie J. Davies, M.D., F.R.C.P.,
Joanne F. Keenan, B.A., Sanjoy Paul, Ph.D., and Jonathan C. Levy, M.D., F.R.C.P.,
for the 4-T Study Group*

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ORIGINAL ARTICLE

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ABSTRACT

BACKGROUND

Adding insulin to oral therapy in type 2 diabetes mellitus is customary when glycemic control is suboptimal, though evidence supporting specific insulin regimens is limited.

METHODS

In an open-label, controlled, multicenter trial, we randomly assigned 708 patients with a suboptimal glycated hemoglobin level (7.0 to 10.0%) who were receiving maximally tolerated doses of metformin and sulfonylurea to receive biphasic insulin aspart twice daily, prandial insulin aspart three times daily, or basal insulin detemir once daily (twice if required). Outcome measures at 1 year were the mean glycated hemoglobin level, the proportion of patients with a glycated hemoglobin level of 6.5% or less, the rate of hypoglycemia, and weight gain.

RESULTS

At 1 year, mean glycated hemoglobin levels were similar in the biphasic group (7.3%) and the prandial group (7.2%) ($P=0.08$) but higher in the basal group (7.6%, $P<0.001$ for both comparisons). The respective proportions of patients with a glycated hemoglobin level of 6.5% or less were 17.0%, 23.9%, and 8.1%; respective mean numbers of hypoglycemic events per patient per year were 5.7, 12.0, and 2.3; and respective mean weight gains were 4.7 kg, 5.7 kg, and 1.9 kg. Rates of adverse events were similar among the three groups.

CONCLUSIONS

A single analogue-insulin formulation added to metformin and sulfonylurea resulted in a glycated hemoglobin level of 6.5% or less in a minority of patients at 1 year. The addition of biphasic or prandial insulin aspart reduced levels more than the addition of basal insulin detemir but was associated with greater risks of hypoglycemia and weight gain. (Current Controlled Trials number, ISRCTN51125379.)

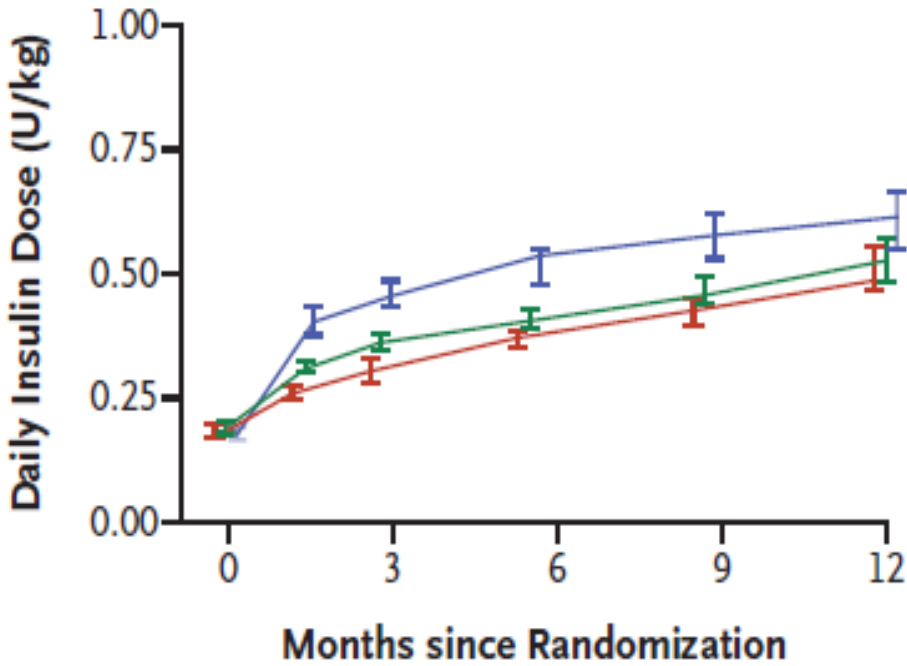
From the Diabetes Trials Unit, Oxford Centre for Diabetes, Endocrinology, and Metabolism (R.R.H., K.I.T., A.J.F., J.F.K., S.P., J.C.L.) and the Department of Primary Health Care (A.J.F.), University of Oxford, Oxford; and the Department of Cardiovascular Sciences, University of Leicester, Leicester (M.J.D.) — all in the United Kingdom. Address reprint requests to Dr. Holman at the Diabetes Trials Unit, Oxford Centre for Diabetes, Endocrinology, and Metabolism, Churchill Hospital, Headington, Oxford OX3 7LJ, United Kingdom, or at rury.holman@dtu.ox.ac.uk.

*Investigators in the Treating to Target in Type 2 Diabetes (4-T) study group are listed in the Appendix.

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Biphasic insulin Prandial insulin Basal insulin



Variable	Biphasic Insulin (N = 235)	Prandial Insulin (N = 239)	Basal Insulin (N = 234)
Insulin dose			
U/day			
Median	48	56	42
U/day/kg			
Median	0.53	0.61	0.49



CELSE GORGORY

HbA1c 9,6 %

Categoría de riesgo **MUY ALTO**

IMC: 37,2 Kg / m²

eTFG E IIIA / V A2

3 ADOs Dosis Máxima

Necesita terapia inyectable para reducir A1C

Considere GLP-1 RA o GIP/GLP-1 RA en la mayoría de las personas antes de la insulina.

INICIACIÓN: Iniciar la dosis inicial adecuada para el agente seleccionado (varía dentro de la clase)

TITULACIÓN: Titular hasta la dosis de mantenimiento (varía dentro de la clase)

Si ya está en GLP-1 RA o GIP dual y GLP-1 RA o si estos no son Apropriado. Se prefiere la insulina.

Si está por encima del objetivo de A1C

AÑADIR INSULINA BASAL

La elección de la insulina basal debe basarse en consideraciones específicas de la persona, incluido el costo. Considere la prescripción de glucagón para Hipoglucemia emergente.

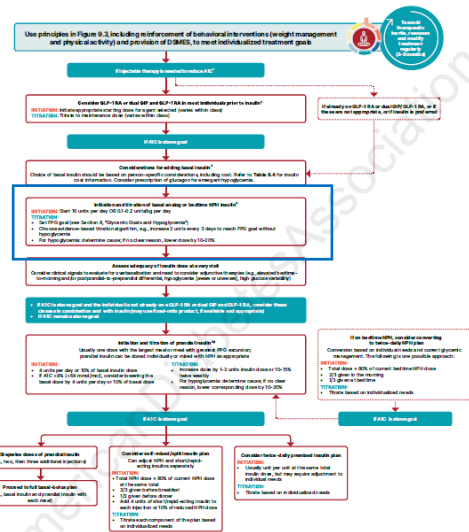
AÑADIR INSULINA BASAL ANÁLOGA O NPH A LA HORA DE ACOSTARSE

INICIACIÓN: Comience 10 unidades por día O 0.1–0.2 unidades/kg por día

TITULACIÓN: Establezca el objetivo glucémicos.

Elija un algoritmo de titulación basado en la evidencia, para alcanzar el objetivo de FPG sin hipoglucemia. Para la hipoglucemia, determine la causa, si no hay una razón clara, reduzca la dosis en un 10-20%

¿CON CUÁL INSULINA BASAL?





The Treat-to-Target Trial

Randomized addition of glargine or human NPH insulin to oral therapy of type 2 diabetic patients

MATTHEW C. RIDDLE, MD¹
JULIO ROSINSTEIN, MD²
JOHN GIBICH, MD³

ON BEHALF OF THE INSULIN GLARGINE 4002
STUDY INVESTIGATORS*

OBJECTIVE — To compare the abilities and associated hypoglycemia risks of insulin glargine and human NPH insulin added to oral therapy of type 2 diabetes to achieve 7% HbA_{1c}.

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RESULTS — Mean FPG at end point was similar with glargine and NPH (117 vs. 120 mg/dl [6.5 vs. 6.7 mmol/l]), as was HbA_{1c} (6.96 vs. 6.97%). A majority of patients (~60%) attained HbA_{1c} ≤7% with each insulin type. However, nearly 25% more patients attained this without documented nocturnal hypoglycemia (≤72 mg/dl [4.0 mmol/l]) with glargine (33.2 vs. 26.7%, *P* < 0.05). Moreover, rates of other categories of symptomatic hypoglycemia were 21–48% lower with glargine.

CONCLUSIONS — Systematically titrating bedtime basal insulin added to oral therapy can safely achieve 7% HbA_{1c} in a majority of overweight patients with type 2 diabetes with HbA_{1c} between 7.5 and 10.0% on oral agents alone. In doing this, glargine causes significantly less nocturnal hypoglycemia than NPH, thus reducing a leading barrier to initiating insulin. This simple regimen may facilitate earlier and effective insulin use in routine medical practice, improving achievement of recommended standards of diabetes care.

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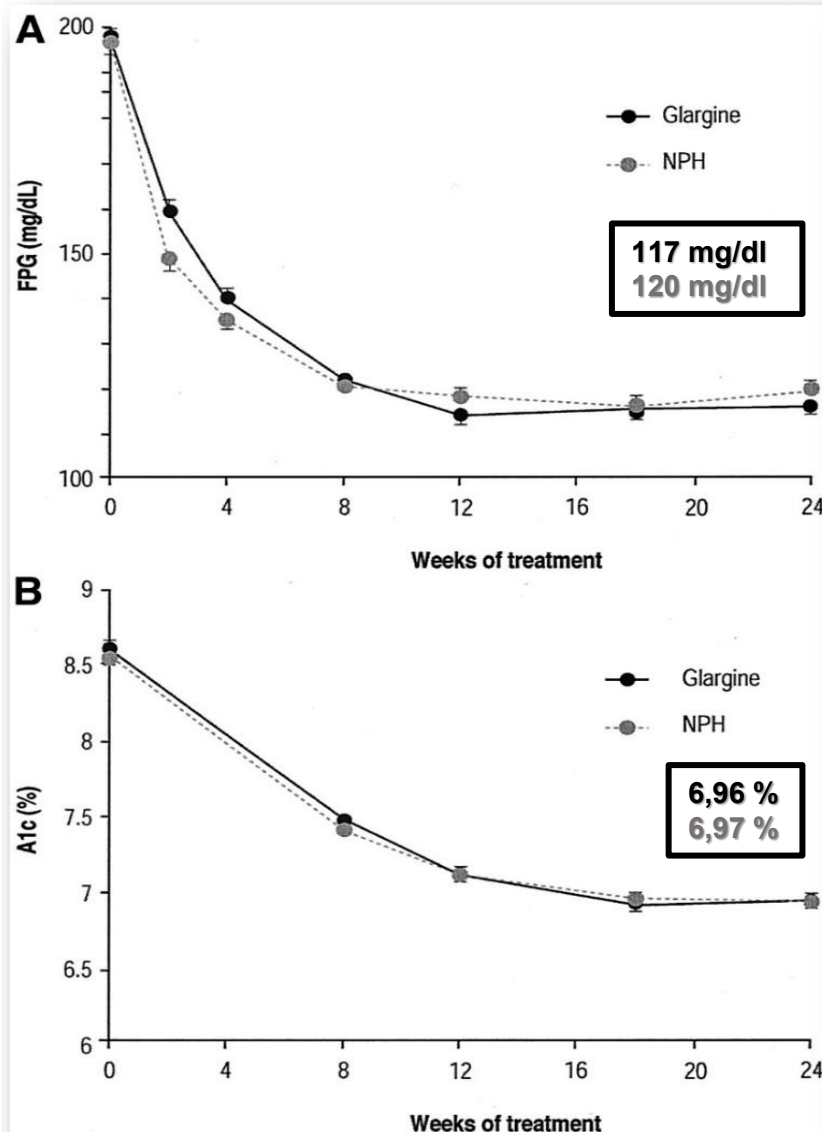
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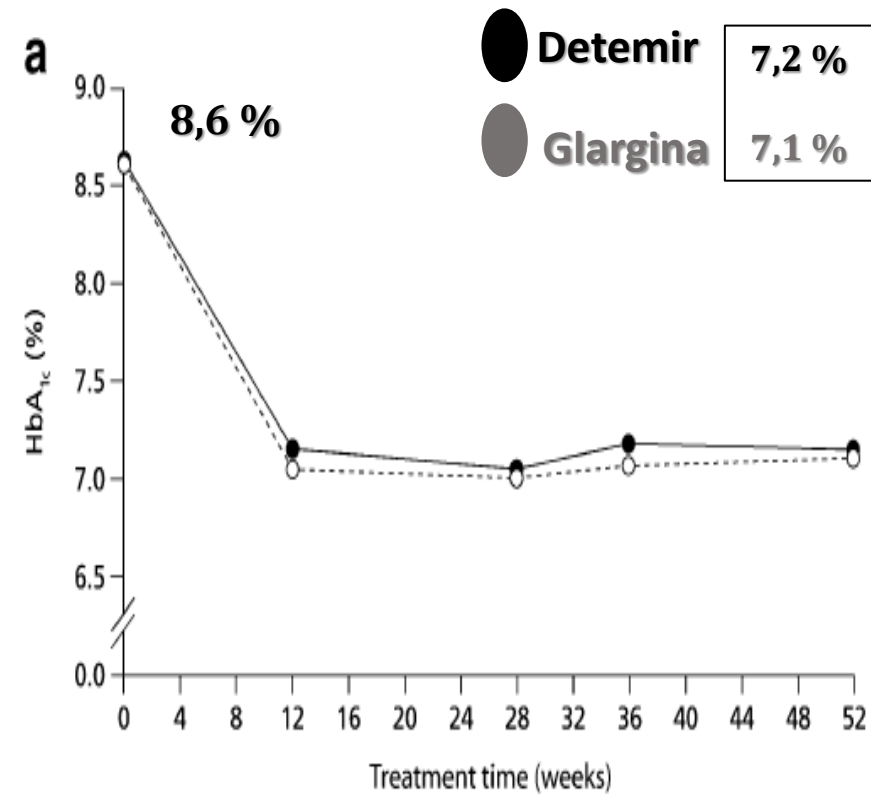
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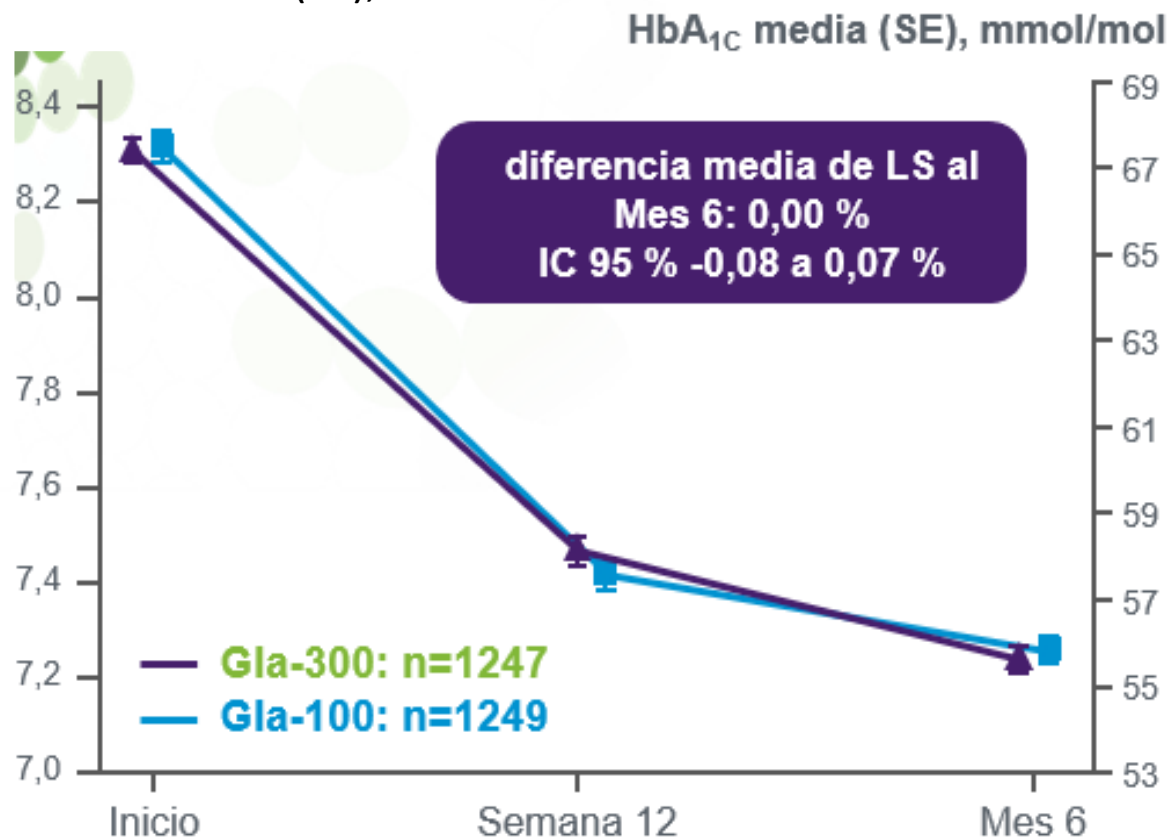
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EDITION TRIALS

GLARGINA U100 Vs U300

HbA1c MEDIA (SE), %





More Similarities Than Differences Testing Insulin Glargine 300 Units/mL Versus Insulin Degludec 100 Units/mL in Insulin-Naive Type 2 Diabetes: The Randomized Head-to-Head BRIGHT Trial

<https://doi.org/10.2337/dci-18-0559>

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BRIGHT (NCT02738151) was a multicenter, open-label, active-controlled, two-arm, parallel-group, 24-week, noninferiority study in insulin-naïve patients with uncontrolled type 2 diabetes. Participants were randomized 1:1 to evening dosing with Gla-300 (N = 466) or IDeg-100 (N = 463), titrated to fasting self-monitored plasma glucose of 80–100 mg/dL. The primary end point was HbA_{1c} change from baseline to week 24. Safety end points included incidence and event rates of hypoglycemia.

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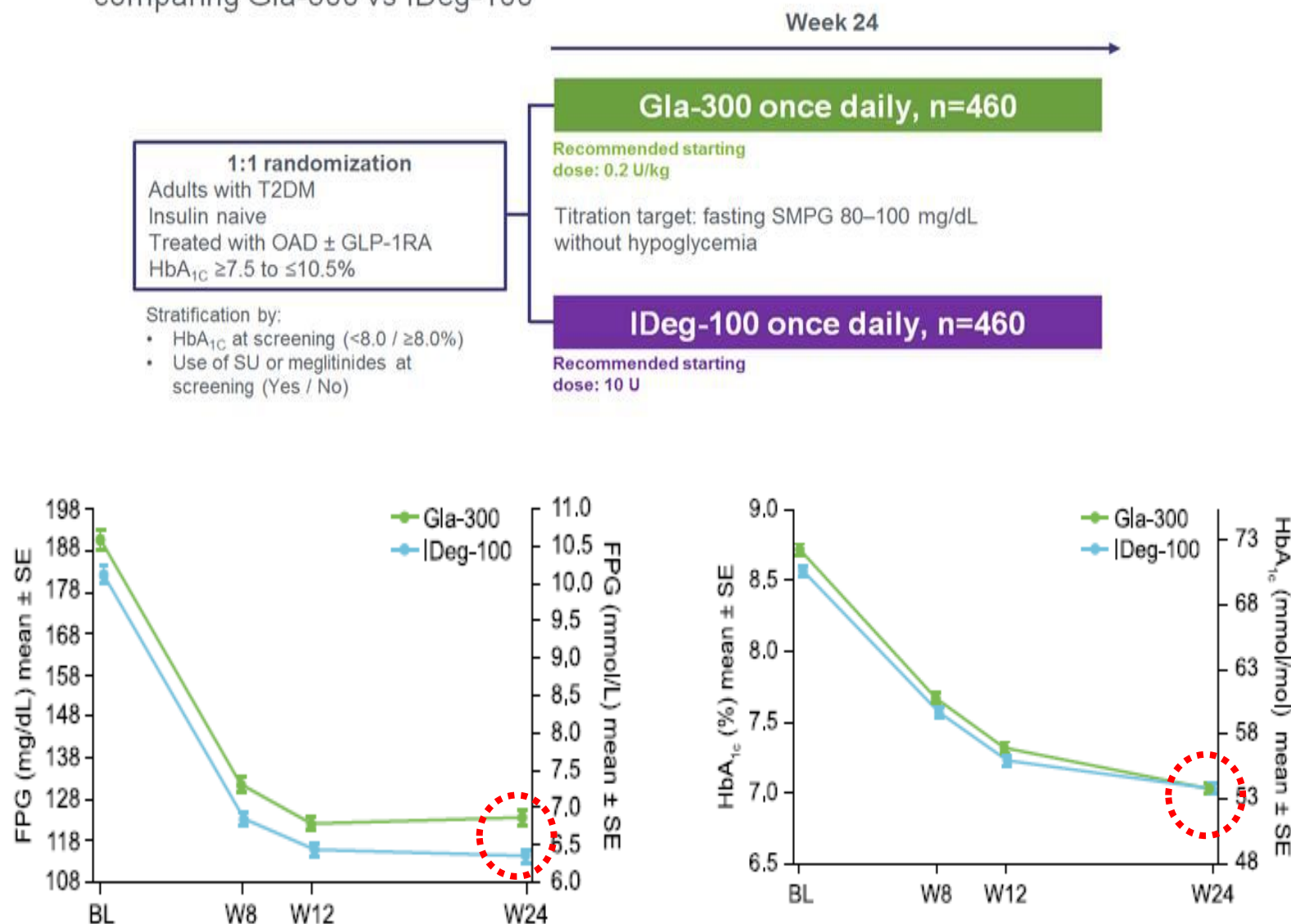
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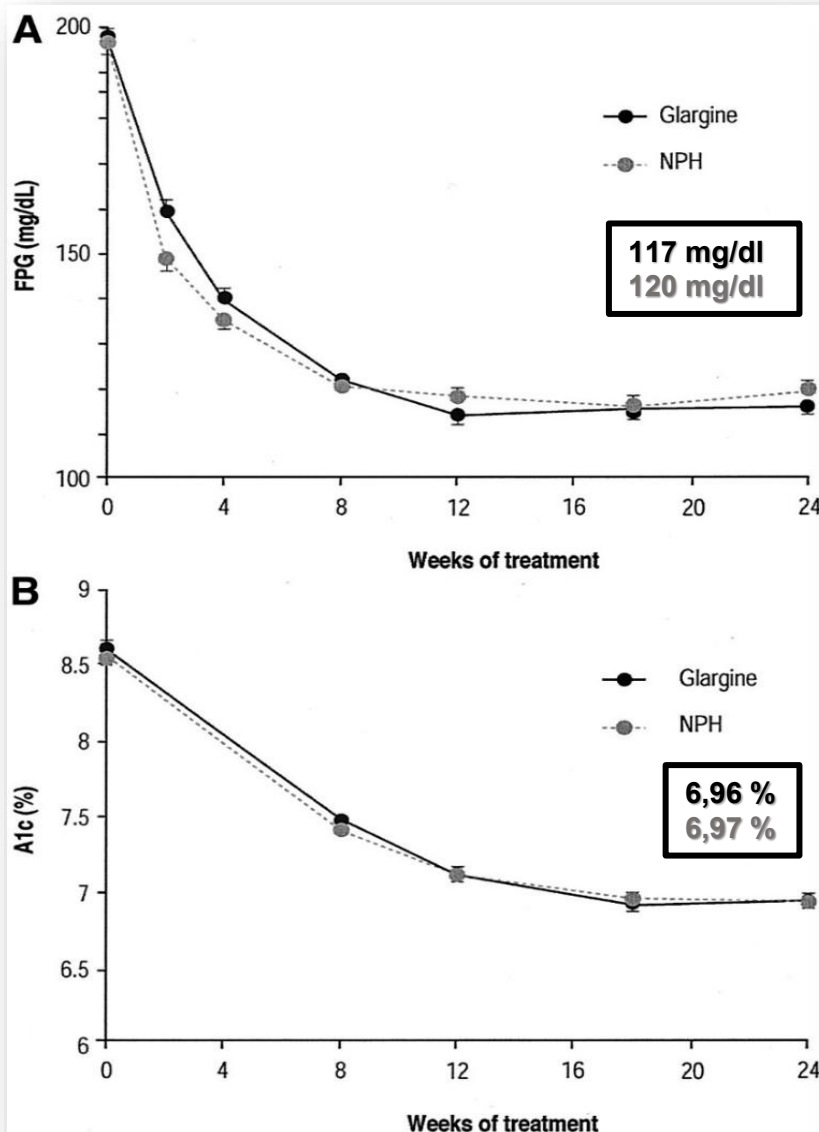
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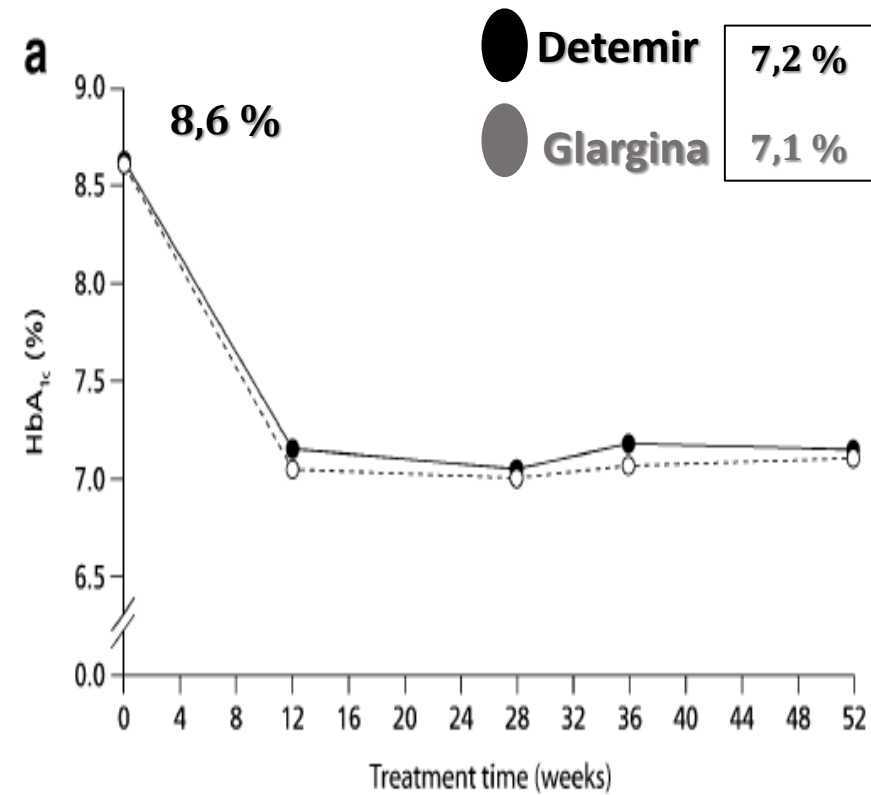
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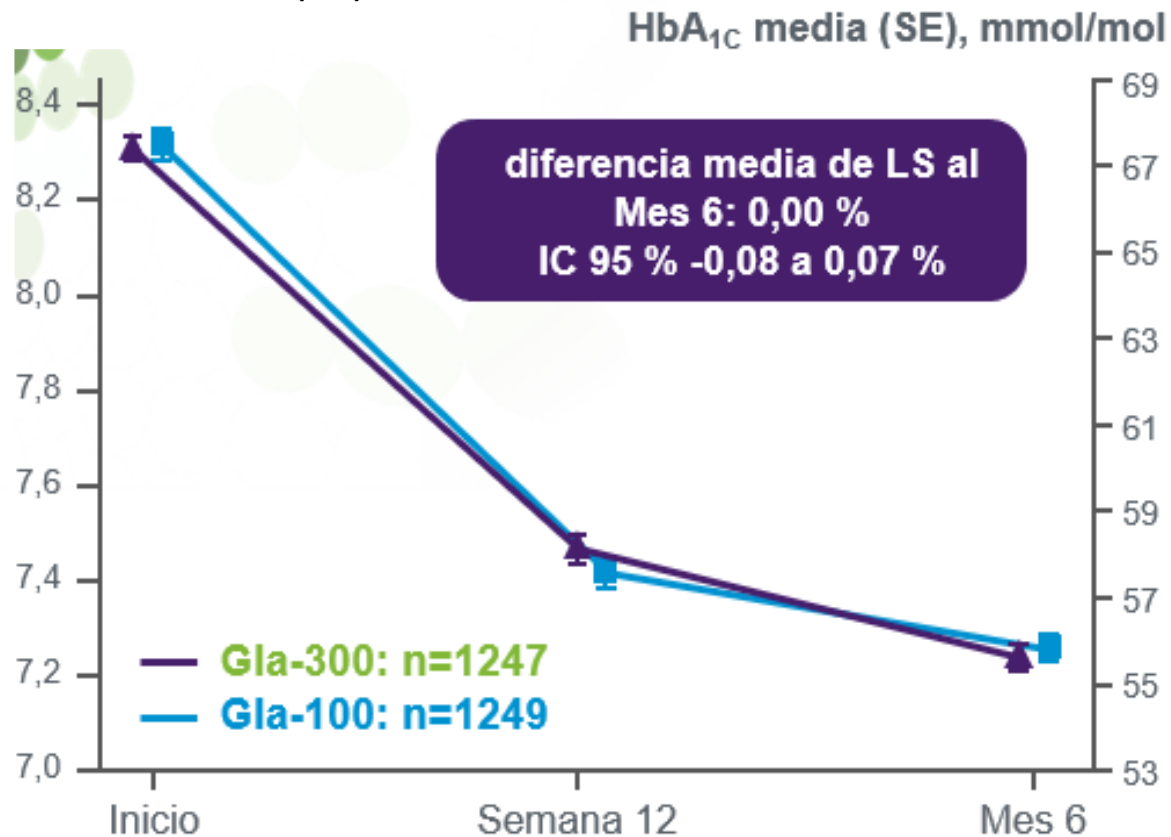


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HbA1c MEDIA (SE), %



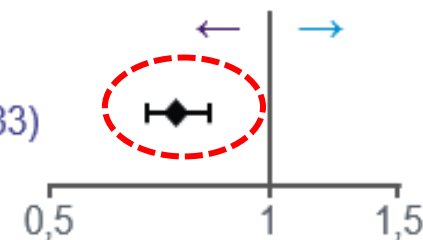
Pacientes con ≥ 1 evento de hipoglucemia confirmada (≤ 70 mg/dl [$\leq 3,9$ mmol/l]) o intensa al Mes 6



Nocturna
(00:00 a 05:59 h)

Favorece a Gla-300 Favorece a Gla-100

0,75 (0,68 a 0,83)



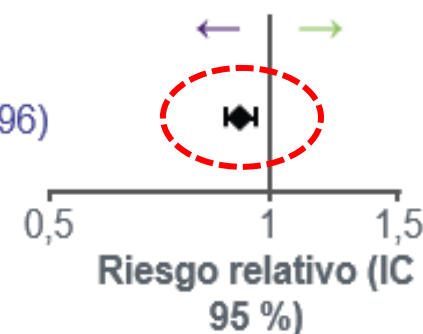
25 %



Cualquier hora
del día (24 h)

Favorece a Gla-300 Favorece a Gla-100

0,91 (0,87 a 0,96)



9 %



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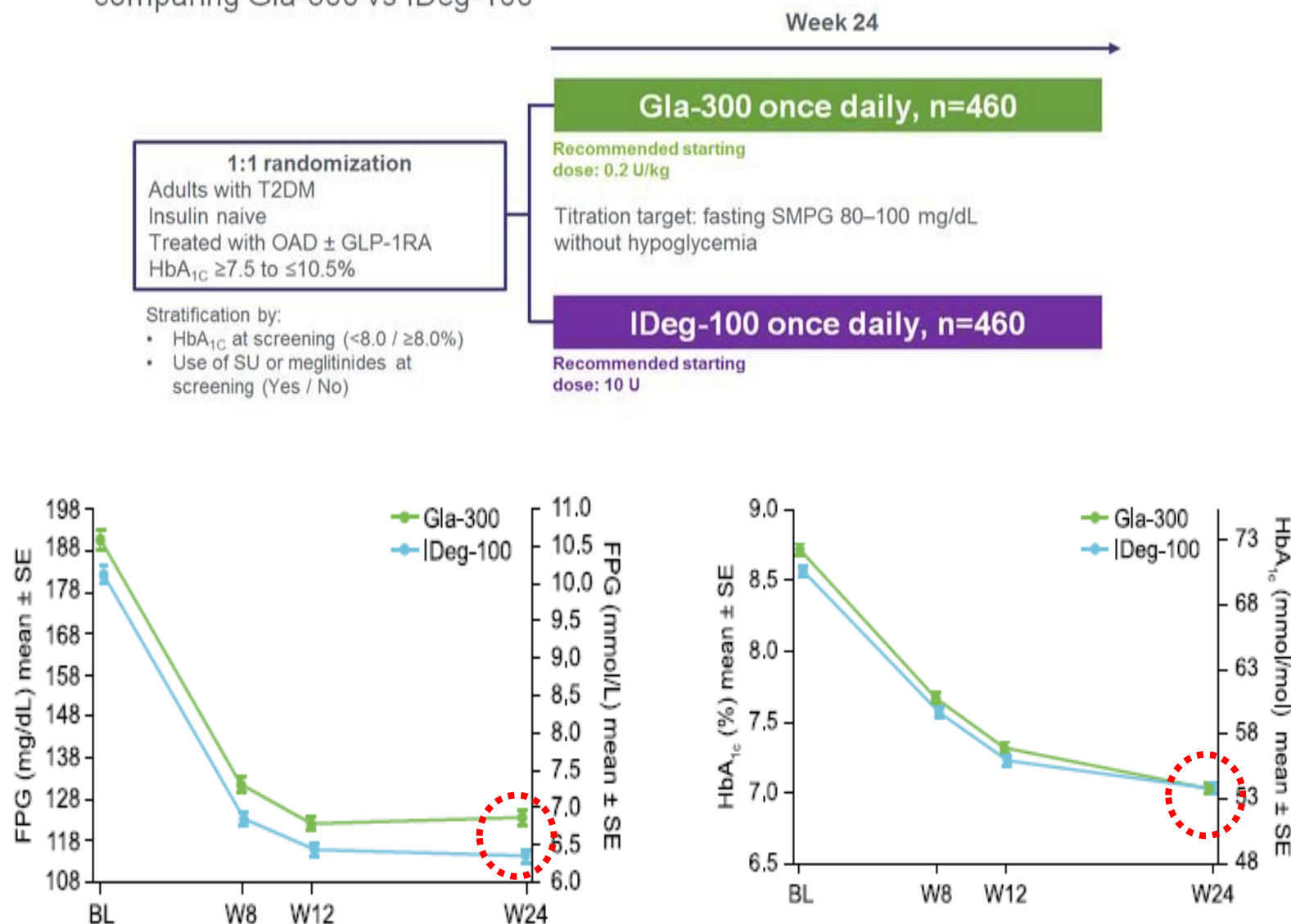
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OBJECTIVE

To compare insulin glargine 300 units/mL (Gla-300) versus insulin degludec 100 units/mL (IDeg-100) in this first head-to-head randomized controlled trial.

RESEARCH DESIGN AND METHODS

BRIGHT (NCT02738151) was a multicenter, open-label, active-controlled, two-arm, parallel-group, 24-week, noninferiority study in insulin-naïve patients with uncontrolled type 2 diabetes. Participants were randomized 1:1 to evening dosing with Gla-300 ($N = 466$) or IDeg-100 ($N = 463$), titrated to fasting self-monitored plasma glucose of 80–100 mg/dL. The primary end point was HbA_{1c} change from baseline to week 24. Safety end points included incidence and event rates of hypoglycemia.

RESULTS

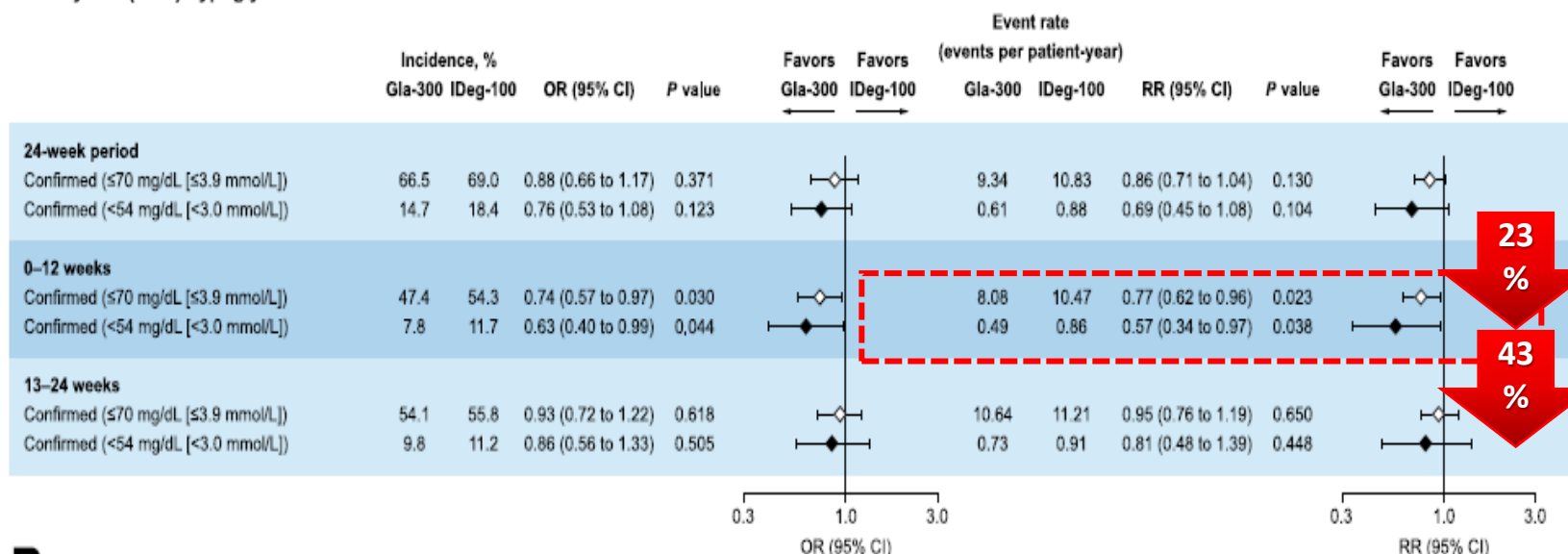
At week 24, HbA_{1c} improved similarly from baseline values of 8.7% (72 mmol/mol) in the Gla-300 group and 8.8% (70 mmol/mol) in the IDeg-100 group to 7.0% (53 mmol/mol)—least squares mean difference -0.05% (95% CI -0.15 to 0.05) (-0.6 mmol/mol [-1.7 to 0.6])—demonstrating noninferiority of Gla-300 versus IDeg-100 ($P < 0.0001$). Hypoglycemia incidence and event rates over 24 weeks were comparable with both insulins, whereas during the active titration period (0–12 weeks) the incidence and rate of anytime (24-h) confirmed hypoglycemia (≤ 70 and < 54 mg/dL) were lower with Gla-300. Both insulins were properly titrated and exhibited no specific safety concerns.

CONCLUSIONS

Gla-300 and IDeg-100 provided similar glycemic control improvements with relatively low hypoglycemia risk. Hypoglycemia incidence and rates were comparable with both insulins during the full study period but lower in favor of Gla-300 during the titration period. The choice between these longer-acting basal insulins may be determined by factors such as access and cost, alongside clinical considerations.

Diabetes Care Publish Ahead of Print, published online August 13, 2018

A Anytime (24-h) hypoglycemia



B

Continuous glucose monitoring-based time-in-range using insulin glargine 300 units/ml versus insulin degludec 100 units/ml in type 1 diabetes: The head-to-head randomized controlled InRange trial

Tadej Battelino MD¹ | Thomas Danne MD² | Steve V. Edelman MD³ |
Pratik Choudhary MD⁴ | Eric Renard MD⁵ | Jukka Westerbacka MD⁶ |
Bhaswati Mukherjee MD⁶ | Valerie Florget MD⁶ | Mathieu Coudert MSc⁶ |
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⁶Sanoft, Paris, France

⁷International Diabetes Center, HealthPartners Institute, Minneapolis

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Funding Information
Sanoft

Abstract

Aims: To use continuous glucose monitoring (CGM)-based time-in-range (TIR) as a primary efficacy endpoint to compare the second-generation basal insulin (BI) analogues insulin glargine 300 U/ml (Gla-300) and insulin degludec 100 U/ml (IDeg-100) in adults with type 1 diabetes (T1D).

Materials and Methods: InRange was a 12-week, multicentre, randomized, active-controlled, parallel-group, open-label study comparing glucose TIR and variability between Gla-300 and IDeg-100 using blinded 20-day CGM profiles. The inclusion criteria consisted of adults with T1D treated with multiple daily injections, using BI once daily and rapid-acting insulin analogues for at least 1 year, with an HbA1c of 7% or higher and of 10% or less at screening.

Results: Overall, 343 participants were randomized: 172 received Gla-300 and 171 IDeg-100. Non-inferiority (10% relative margin) of Gla-300 versus IDeg-100 was shown for the primary endpoint (percentage TIR ≥ 70 to ≤ 180 mg/dL) least squares (LS) mean (95% confidence interval) 52.74% (51.06%, 54.42%) for Gla-300 and 55.09% (53.34%, 56.84%) for IDeg-100; LS mean difference (non-inferiority): 3.16% (0.88%, 5.44%) (non-inferiority $P = .0067$). Non-inferiority was shown on glucose total coefficient of variation (main secondary endpoint): LS mean 39.93% (39.20%, 40.61%) and 41.22% (40.49%, 41.95%), respectively; LS mean difference (non-inferiority) -5.44% (-6.50% , -4.38%) (non-inferiority $P < .0001$). Superiority of Gla-300 over IDeg-100 was not shown on TIR. Occurrences of self-measured and CGM-derived hypoglycaemia were comparable between treatment groups. Safety profiles were consistent with known profiles, with no unexpected findings. **Conclusions:** Using clinically relevant CGM metrics, InRange shows that Gla-300 is non-inferior to IDeg-100 in people with T1D, with comparable hypoglycaemia and safety profiles.

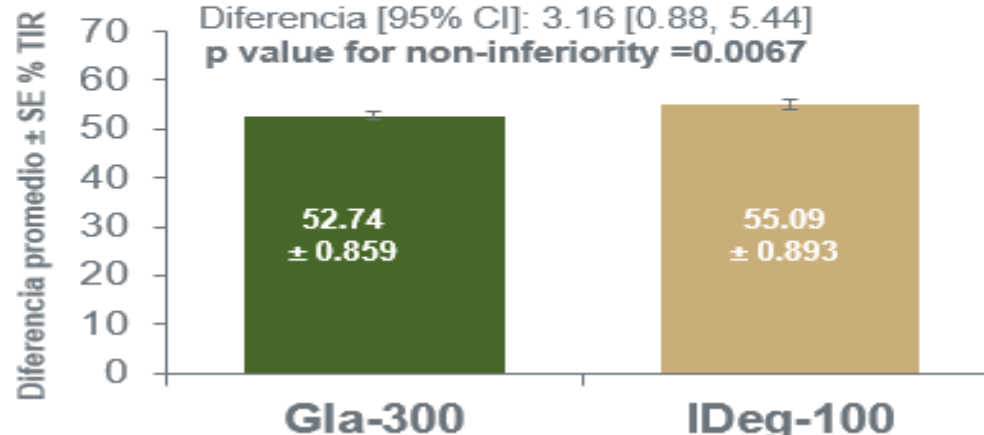
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Desenlace primario TIEMPO EN RANGO

% TIR ≥ 70 a ≤ 180 mg/dL a la semana 12

No inferioridad*

Diferencia [95% CI]: 3.16 [0.88, 5.44]
p value for non-inferiority = 0.0067



Continuous glucose monitoring-based time-in-range using insulin glargine 300 units/ml versus insulin degludec 100 units/ml in type 1 diabetes: The head-to-head randomized controlled InRange trial

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⁵Department of Endocrinology, Diabetes and Nutrition, Montpellier University Hospital, University of Montpellier, Montpellier, France

⁶Sanofi, Paris, France

⁷International Diabetes Center, HealthPartners Institute, Minneapolis

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Email: tadej.battelino@umc-lj.si

Funding Information
Sanofi

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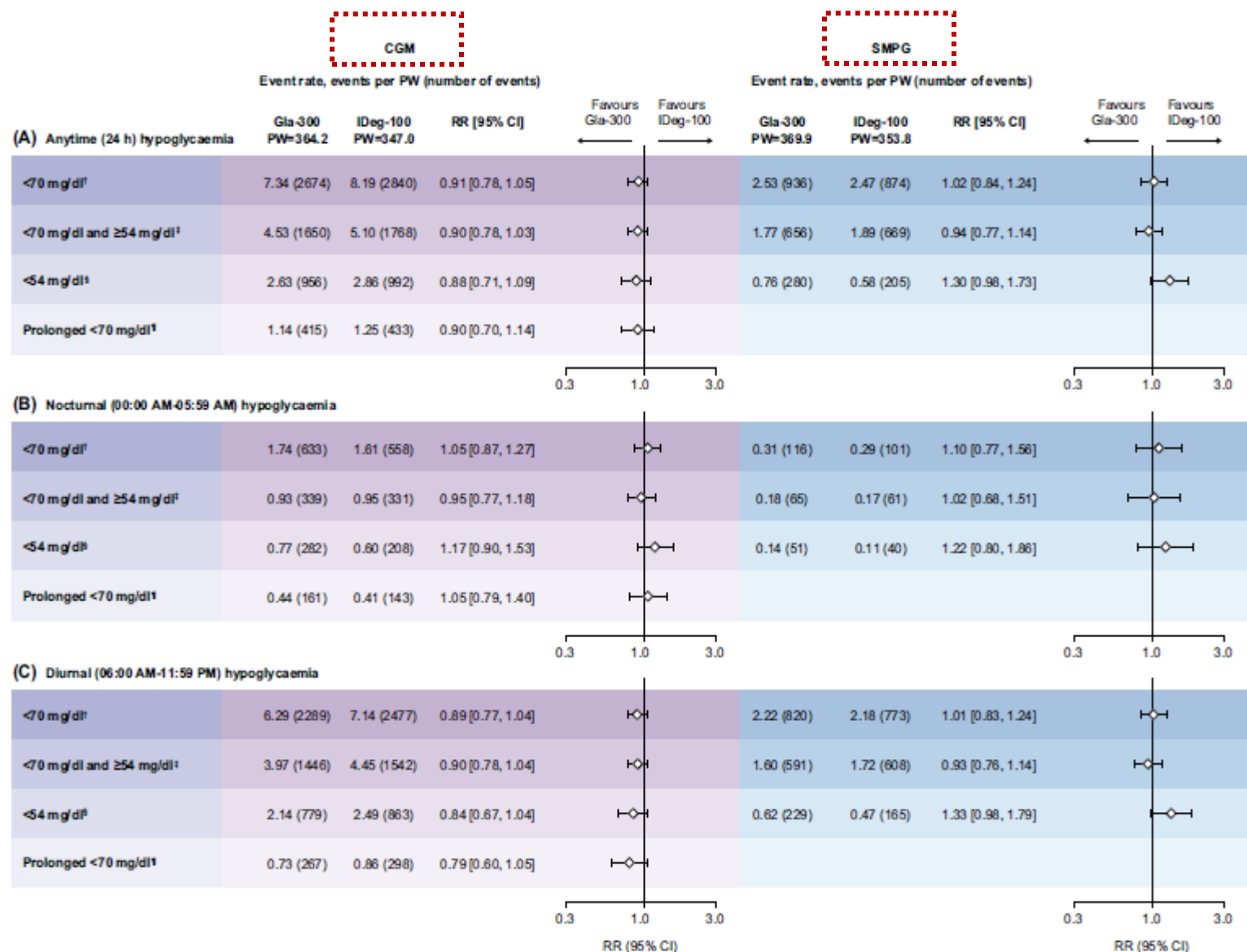
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HIPOGLICEMIA





Sr. CELSO GORGORY

HbA1c 9,6 %

Categoría de riesgo **MUY ALTO**

IMC: 37,2 Kg / m²

eTFG E IIIA / V A2

3 ADOs Dosis Máxima

Necesita terapia inyectable para reducir A1C

Considere GLP-1 RA o GIP/GLP-1 RA en la mayoría de las personas antes de la insulina.

INICIACIÓN: Iniciar la dosis inicial adecuada para el agente seleccionado (varía dentro de la clase)

TITULACIÓN: Titular hasta la dosis de mantenimiento (varía dentro de la clase)

Si ya está en GLP-1 RA o GIP dual y GLP-1 RA o si estos no son Apropriado. Se prefiere la insulina.

Si está por encima del objetivo de A1C

AÑADIR INSULINA BASAL

La elección de la insulina basal debe basarse en consideraciones específicas de la persona, incluido el costo. Considere la prescripción de glucagón para Hipoglucemia emergente.

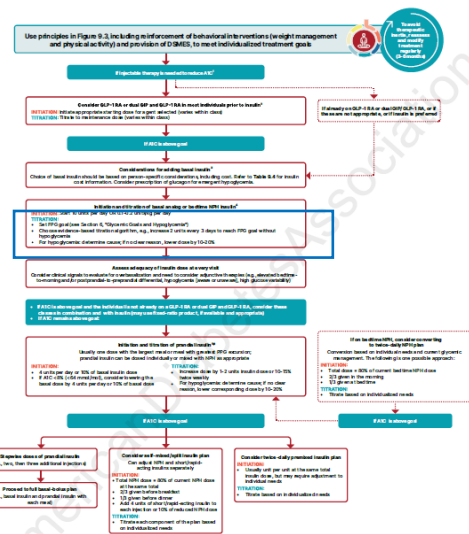
AÑADIR INSULINA BASAL ANÁLOGA O NPH A LA HORA DE ACOSTARSE

INICIACIÓN: Comience 10 unidades por día O 0.1–0.2 unidades/kg por día

TITULACIÓN: Establezca el objetivo glucémicos.

Elija un algoritmo de titulación basado en la evidencia, para alcanzar el objetivo de FPG sin hipoglucemia. Para la hipoglucemia, determine la causa, si no hay una razón clara, reduzca la dosis en un 10-20%

¿CÓMO LO HAGO?



TITULACION Y ESTUDIOS CLINICOS

ESTUDIO	DOSIS DE INICIO	INCREMENTO DE LA DOSIS-	FRECUENCIA	DOSIS FINAL	HBA1C
 LANMET*1. N=206	10UI PACIENTES CON METFORMINA 20 UI PACIENTES CON SULFONILUREAS Y METFORMINA	- >80 MG/DL - 2UI - <130 MG/DL + 2UI	CADA 3 DIAS	66 UI	7,14%
TREAT TO TARGET*2. N=206	10 UI	100-120 MG/DL + 2 UI 120-140 MG/DL +4 UI 140-180 MG/DL +6 UI > 180 MG/DL +8UI	CADA SEMANA	47 UI	6,96%
 INSIGHT*3 N=206	10 UI	>130 mg / dL + 1 UI < 80 mg /dL -1 UI	TODOS LOS DIAS	38 UI	6,96%

- 1. YKI-JARVINEN H, ET AL. DIABETOLOGIA 2006; 46 (3) 442- 4516.
- 2. RIDDLE MC, ROSENSTOCK J, GERICH: DIABETES CARE, 2003; 26(11) 3080-30865.
- 3. GERSTEIN ET AL. DIABET MED.2006; 23; 736-7427 MULLINS P ET AL, CLIN THERAPEUTICS 2007; 29: 1607-1613.



CELSO GORGORY

HbA1c 9,6 %

Categoría de riesgo **MUY ALTO**

IMC: 37,2 Kg / m2 Peso: **105** Kg

eTFG E IIIA / V A2

3 ADOs Dosis Máxima

DOSIS **0,2** UI X KG

20 UI SC C 24H

.....

META

GLUCOMETRICA

(80-130mg/dL)

REGISTRO DE GLUCOMETRIAS

FECHA	GLUCOMETRIA	DOSIS	FECHA	GLUCOMETRIA	DOSIS	FECHA	GLUCOMETRIA	DOSIS
LUNES	180	20	LUNES	130	26	LUNES	108	26
MARTES	168	20	MARTES	115	26	MARTES	96	26
MIERCOLES	175	20	MIERCOLES	110	26	MIERCOLES	110	26
JUEVES	150	22	JUEVES	98	26	JUEVES	115	26
VIERNES	152	22	VIERNES	72	26	VIERNES	90	26
SABADO	148	22	SABADO	78	26	SABADO	137	26
DOMINGO	135	24	DOMINGO	80	28	DOMINGO	140	26
LUNES	140	24	LUNES	70	28	LUNES	115	28
MARTES	138	24	MARTES	72	28	MARTES	110	28
MIERCOLES	130	24	MIERCOLES	68	26	MIERCOLES	112	28
JUEVES	135	24	JUEVES	85	26	JUEVES	88	28
VIERNES	135	24	VIERNES	78	26	VIERNES	95	28
SABADO	128	26	SABADO	97	26	SABADO	105	28
DOMINGO	125	26	DOMINGO	110	26	DOMINGO	118	28



CELSO GORGORY

HbA1c 9,6 %

Categoría de riesgo **MUY ALTO**

IMC: 37,2 Kg / m²

eTFG E IIIA / V A2

3 ADOs Dosis Máxima

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AÑADIR INSULINA BASAL ANÁLOGA O NPH A LA HORA DE ACOSTARSE

INICIACIÓN: Comience 10 unidades por día O 0.1–0.2 unidades/kg por día

TITULACIÓN: Establezca el objetivo glucémicos.

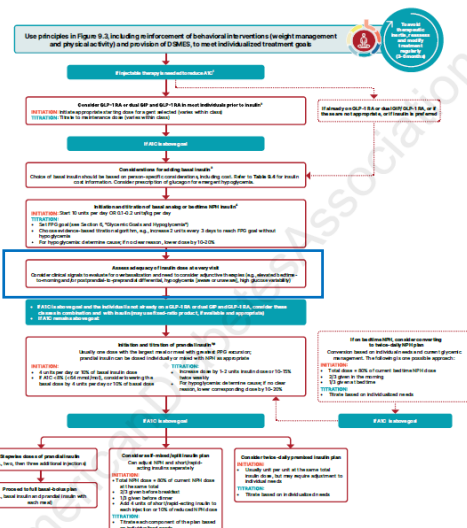
Elija un algoritmo de titulación basado en la evidencia, para alcanzar el objetivo de FPG sin hipoglucemia. Para la hipoglucemia, determine la causa, si no hay una razón clara, reduzca la dosis en un 10-20%

EVALUAR LA IDONEIDAD DE LA DOSIS DE INSULINA BASAL

Considere las señales clínicas para evaluar la sobrebasalización y la necesidad de considerar terapias complementarias (p. ej., dosis basal superior a ~0,5 unidades/kg/día, dosis elevada diferencial hora de acostarse-mañana y/o posprandial, hipoglucemia [consciente o inconsciente], alta variabilidad)

- Si está por encima del objetivo de A1C y aún no está en un GLP-1 RA o GIP dual y GLP-1 RA, considere estas clases, ya sea en combinación libre o combinación de proporción fija, con insulina.

¿HASTA QUE DOSIS ?



ORIGINAL ARTICLE

WILEY

When basal insulin is not enough: A dose-response relationship between insulin glargine 100 units/mL and glycaemic control

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Funding information

This study was funded by Sanofi US, Inc.

Aims: A post-hoc analysis to assess the impact in people with type 2 diabetes, of increasing doses of basal insulin on glycaemic measures, body weight and hypoglycaemia.

Research Design and Methods: We included data from prospective, randomized controlled treat-to-target trials of ≥ 24 weeks' duration in people with type 2 diabetes, uncontrolled on metformin and sulphonylureas, and treated with insulin glargine 100 units/mL (U100), who had at least six fasting plasma glucose (FPG) measurements. The impact of insulin dose on glycated haemoglobin (HbA1c) values, FPG, hypoglycaemia incidence (<3.9 mmol/L [70 mg/dL]), and body weight was analysed. A total of 458 participants from three eligible trials were included.

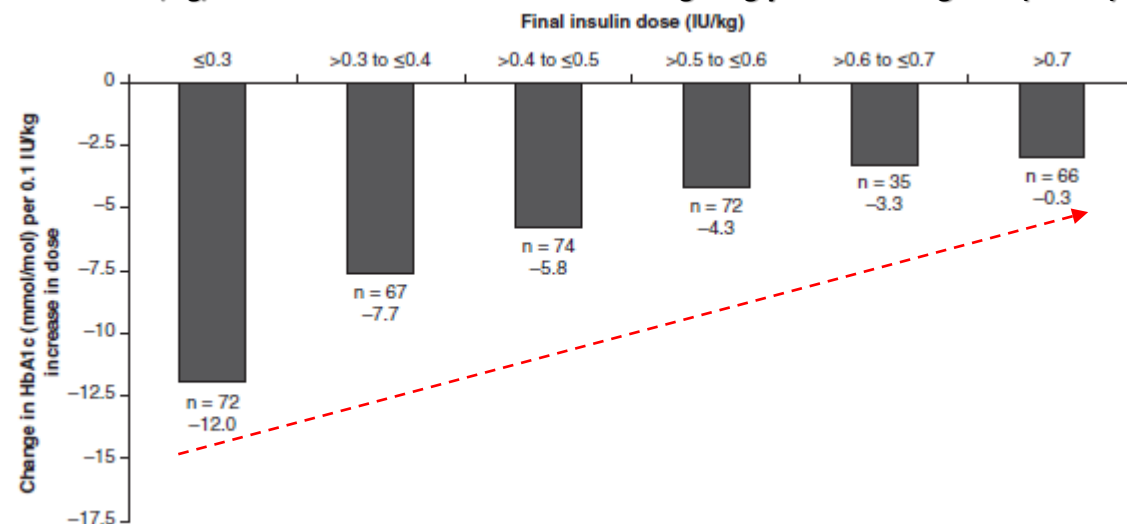
Results: The observed relationship between higher basal insulin doses and glycaemic control was non-linear, with increasing insulin dose leading to smaller reductions in FPG and HbA1c for doses >0.3 IU/kg/d, with a plateauing effect at 0.5 IU/kg/d. Total daily dose of insulin >0.5 IU/kg/d resulted in greater weight gain, but without higher rates of hypoglycaemia, compared with insulin doses ≤ 0.5 IU/kg/d.

Conclusions: This analysis indicates that basal insulin doses >0.5 IU/kg/d have diminishing additional impact on improving glycaemic measures, with the disadvantage of additional weight gain. Clinicians should consider anti-hyperglycaemic treatment intensification at doses approaching 0.5 IU/kg/d.

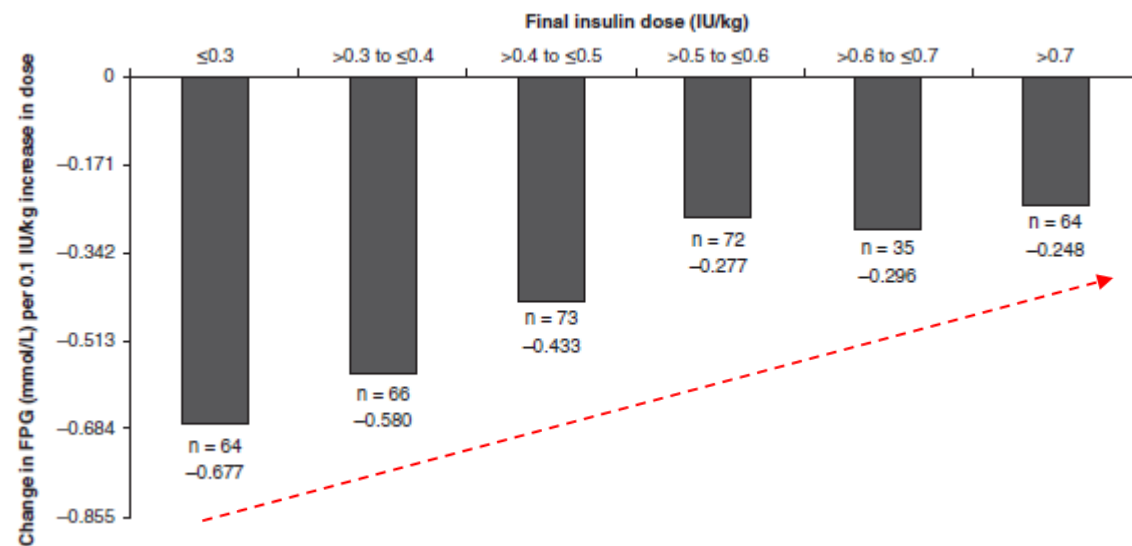
KEYWORDS

basal insulin, glargine, HbA1c, postprandial, type 2 diabetes

Effect of 0.1-IU/kg/d increases in insulin dose on the change in glycated haemoglobin (HbA1c)



Effect of 0.1-IU/kg/d increases in daily insulin dose on the change in fasting plasma glucose (FPG)



When basal insulin is not enough: A dose-response relationship between insulin glargine 100 units/mL and glycaemic control

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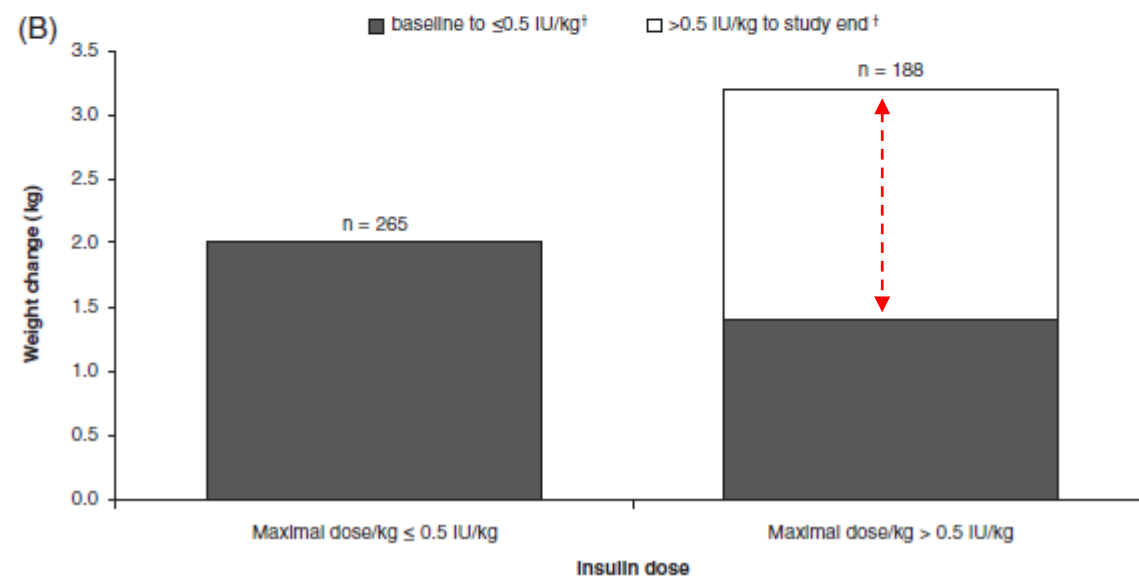
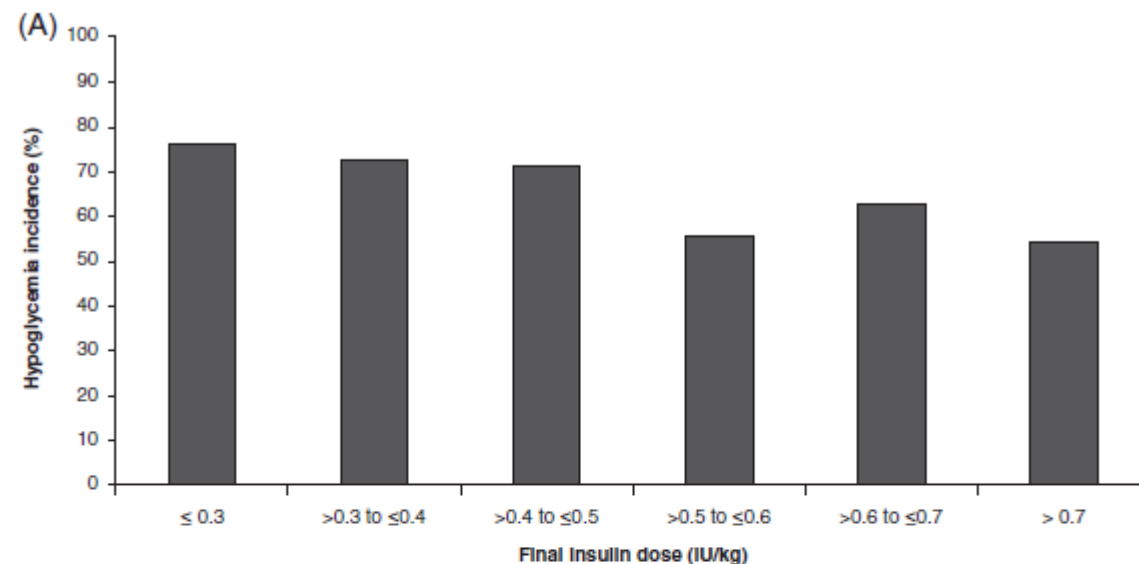
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KEYWORDS

basal insulin, glargine, HbA1c, postprandial, type 2 diabetes



A, Percentage of participants reporting hypoglycaemia (< 3.9 mmol/L [70 mg/dL]) for each 0.1-IU/kg/d dose increment. B, Weight change from baseline to study end by daily basal insulin dose (maximal dose/kg/day). [†]For those patients with maximal dose of > 0.5 IU/kg/d (right-hand bar), the graph is stratified by weight increase from baseline to when the dose reached 0.5 IU/kg/d (dark bar), and then the increase from > 0.5 IU/kg/d to study end (light bar)

ORIGINAL PAPER

IJCP THE INTERNATIONAL JOURNAL OF
CLINICAL PRACTICE

How much is too much? Outcomes in patients using high-dose insulin glargine

T. Reid,¹ L. Gao,² J. Gill,³ A. Stuhr,³ L. Traylor,³ A. Vlajnic,³ A. Rhinehart⁴

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Disclosure:
Timothy Reid and Ghazal
Rider, at Novartis, Sanofi-
Aventis, Amgen, Eli Lilly,
Ingham, and Lilly, are
consultants for Sanofi, Inc.
Gill, Stuh, Traylor, and
Vlijmic are employees of
Novartis, Inc.; and
Rhinehart is a consultant for
Novartis, Sanofi, and
Amgen. Vlijmic is also a
consultant for Sanofi,
Novartis, and Amgen.

SUMMARY

Background and objectives: Many patients with type 2 diabetes mellitus (T2DM) do not achieve glycaemic control targets on basal insulin regimens. This analysis investigated characteristics, clinical outcomes and impact of concomitant oral antidiabetic drugs (OADs) in patients with T2DM treated with high-dose insulin glargine. **Methods:** Patient-level data were pooled from 15 randomised, treatment-to-target trials in patients with T2DM treated with insulin glargine $\pm$ OADs for ≥ 24 weeks. Data were stratified according to whether patients exceeded three insulin dose cut-off levels (> 0.5 , > 0.7 and > 1.0 IU/kg). End-points included glycaemic haemoglobin A1c (A1C), fasting plasma glucose, body weight, and overall, nocturnal and severe hypoglycaemia. **Results:** Data from 2837 insulin-naïve patients were analysed. Patients with insulin titrated beyond the three doses investigated had significantly higher baseline A1C levels and were younger, with shorter diabetes duration than those below cut-offs ($p < 0.05$ for all cut-offs); they also had greater weight gain ($p < 0.001$ for the > 0.5 and > 0.7 IU/kg cut-offs) than those who did not exceed the cut-offs, regardless of concomitant OAD. Patients on concomitant metformin alone had higher insulin doses at Week 24, but achieved greater reductions in A1C, less weight gain and lower hypoglycaemia rates than patients on a concomitant sulfonylurea or metformin plus a sulfonylurea, regardless of whether cut-offs were exceeded. **Conclusion:** In patients with T2DM, increasing basal insulin doses above 0.5 IU/kg may not improve glycaemic control; treatment strategies targeting postprandial glucose control should be considered for such patients.

Introduction

The joint position statement of the American Diabetes Association (ADA) and the European Association for the Study of Diabetes (EASD) notes that, because of disease progression, many patients with type 2 diabetes mellitus (T2DM) ultimately require insulin therapy, either alone or in combination with other oral antidiabetic drugs (OADs), to maintain glycaemic control (1). Guidance from the ADA/EASD and also the American Association of Clinical Endocrinologists suggests that insulin should be initiated as a single daily injection of basal insulin at a dose of 0.1–0.2 IU/kg of body weight although higher doses (e.g. 0.2–0.4 IU/kg) are appropriate for patients who are severely hyperglycaemic (1,2). The ADA/EASD position statement also alerts practitioners to consider the need for additional mealtime insulin or consider a glucagon-like peptide 1

(GLP-1) receptor agonist trial to reduce postprandial hyperglycaemia once the daily insulin dose exceeds 0.5 IU/kg and particularly as it approaches 1.0 IU/kg (1).

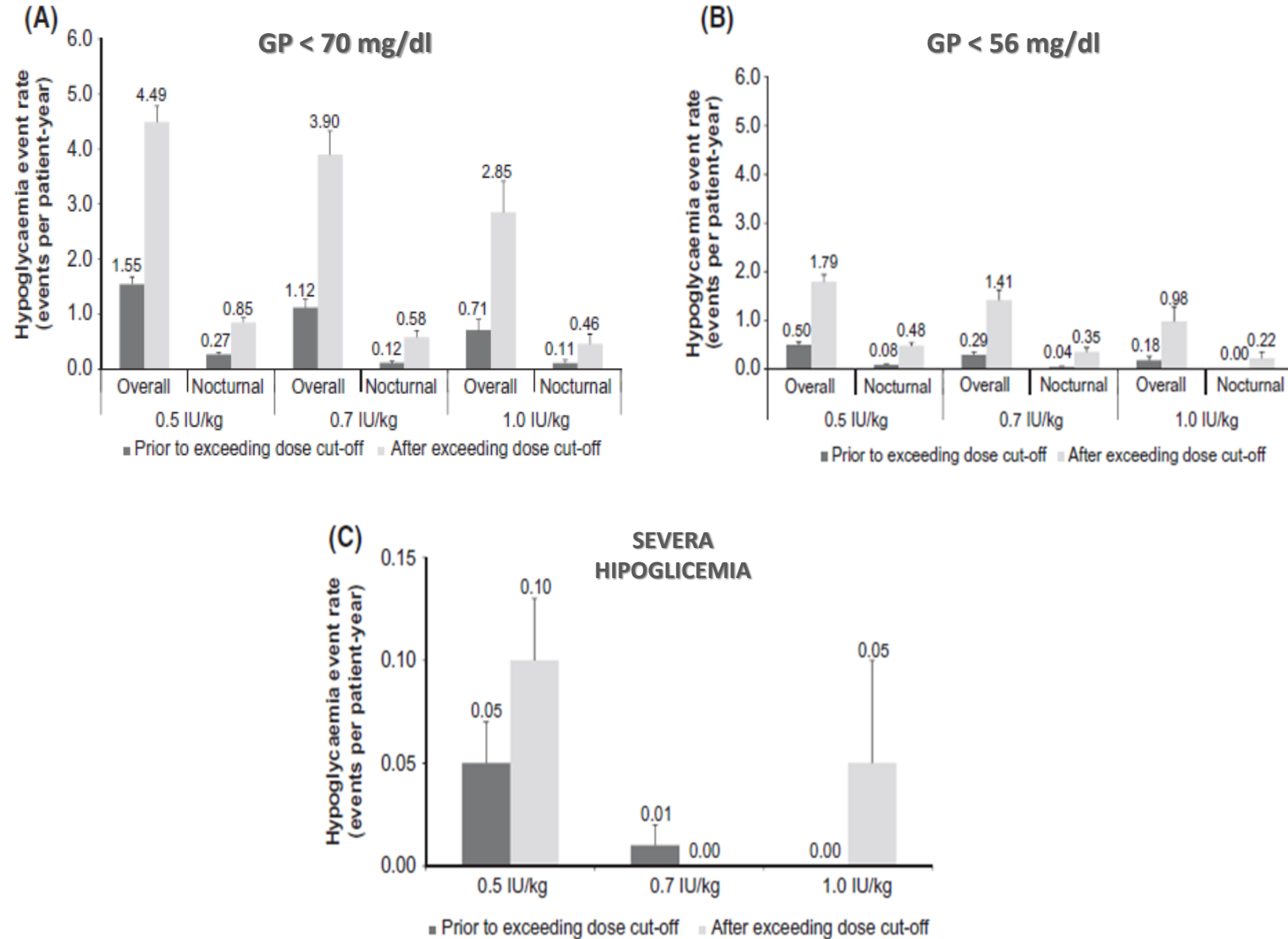
Recent cross-sectional data from the National Health and Nutrition Examination Surveys (NHANES) suggest that the proportion of patients who achieve a glycaemic haemoglobin A1c (A1C) goal of $< 7.0\%$ has increased over the last decade, though only 52.5% of the NHANES participants in the 2007–2010 survey period met the A1C target of $< 7.0\%$ (3). This may be attributed to clinical inertia, defined as failure to intensify diabetes treatment appropriately when the A1C goal of $< 7.0\%$ is not achieved – which affects about one-third of patients with T2DM, including 26.1% of patients on insulin monotherapy and 21.4% of patients receiving insulin plus OADs (4). Clinical inertia is often a consequence of the discrepancy between evidence-based

What's known

A large proportion of patients with type 2 diabetes mellitus do not achieve the glycaemic haemoglobin A1c (A1C) goals of $< 7.0\%$ recommended in current guidelines. This may be attributed to clinical inertia in initiating insulin therapy and the failure to intensify diabetes treatment appropriately when the A1C goal of $< 7.0\%$ is not achieved.

What's new

This analysis highlights that continued upward titration of basal insulin glargine to doses > 0.5 , > 0.7 and even > 1.0 IU/kg does not appear to result in further improvements in glycaemic control but, for those patients needing higher doses, is associated with increased weight gain and also an increase in risk of hypoglycaemia once the dose cut-off is exceeded. Intensification of treatment to control residual postprandial hyperglycaemia should be considered for patients whose diabetes is inadequately controlled with high-dose insulin glargine plus oral antidiabetic drugs.





CASO 2 : INTENSIFICACIÓN EN PERSONAS NO CONTROLADAS ADECUADAMENTE (HbA 1c > 7%) CON INSULINA BASAL

Perfil clínico

Nombre: **BARNEY GÓMEZ**
Edad: 56 años
Duración de la diabetes: 10 años
IMC: 36.6 kg/ m²
Talla 162 CM W: 96Kg



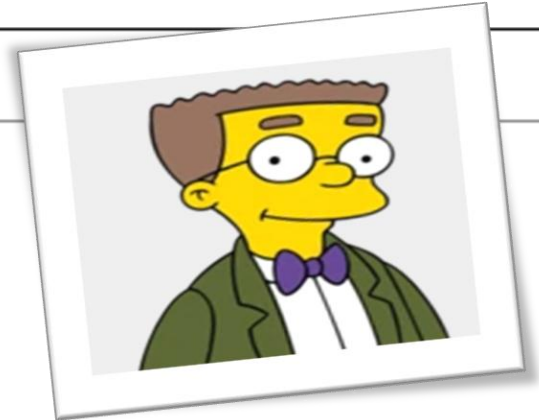
Resultados de laboratorio: HbA 1c : 9,2%; Creatinina: 0,8 mg/dL;
TFGe: 91/mL/min/1,73 m², RAC 18mg/g c-Ldl 68 mg/dL

Antecedentes médicos previos: Consumo excesivo de alcohol, Hipertensión, Dislipidemia, Obesidad 2 hipotiroidismo, MASLD (**metALD**), Episodios hipoglucemia grado 1 nocturnos poco frecuente (<70 mg/dl).

Medicación actual: I.Glar 38 U/día, Perindopril / Amlodipino 5+5mg día, Atorvastatina 40 mg día, levotiroxina 112 mcg día, Duloxetina 30mg día Metformina 1000mg 2v día, Linagliptina 5 mg día, Dapagliflozina 10mg día.

Perfil clínico

Nombre: **WAYLON SMITHERS JR.**
Edad: 56 años
Duración de la diabetes: 10 años
IMC: 23 kg/ m²
Talla 162 CM W: 59,5Kg



Resultados de laboratorio: HbA 1c : 9,2%; Creatinina: 0,8 mg/dL;
TFGe: 91/mL/min/1,73 m², RAC 18mg/g c-Ldl 68 mg/dL

Antecedentes médicos previos: Hipertensión, Dislipidemia, hipotiroidismo, depresión actualmente bien controlada con medicación, hipoglucemia no grave poco frecuente (<70 mg/dl) en el año anterior.

Medicación actual: I.Glar 28 U/día, Perindopril / Amlodipino 5+5mg día, Atorvastatina 40 mg día, levotiroxina 112 mcg día, Duloxetina 30mg día Metformina 1000mg 2v día, Linagliptina 5 mg día, Dapagliflozina 10mg día.

Perfil clínico

Nombre: **BARNEY GÓMEZ**

Edad: 56 años

Duración de la diabetes: 10 años

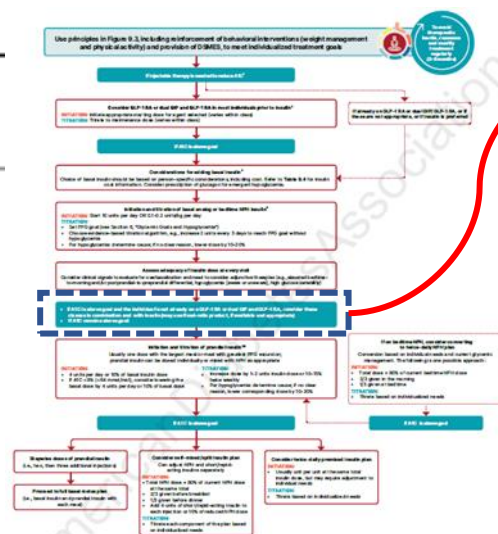
IMC: 36.6 kg/m²

Talla 162 CM W: 96Kg

Resultados de laboratorio: HbA_{1c} : 9,2%; Creatinina: 0,8 mg/dL;
TFGe: 91/mL/min/1,73 m², RAC 18mg/g c-Ldl 68 mg/dL

Antecedentes médicos previos: Hipertensión, Dislipidemia, Obesidad 2
hipotiroidismo, MASLD (**metALD**), Episodios hipoglucemia grado 1
nocturnos poco frecuente (<70 mg/dl).

Medicación actual: I.Glar 38 U/día, Perindopril / Amlodipino 5+5mg día,
Atorvastatina 40 mg día, levotiroxina 112 mcg día, Duloxetine 30mg día
Metformina 1000mg 2v día, Linagliptina 5 mg día, Dapagliflozina 10mg
día.



- Si está por encima del objetivo de A1C y aún no está en un GLP-1 RA o GIP dual y GLP-1 RA, considere estas clases, ya sea en combinación libre o combinación de proporción fija, con insulina.





Efficacy and Safety of IDegLira Versus Basal-Bolus Insulin Therapy in Patients With Type 2 Diabetes Uncontrolled on Metformin and Basal Insulin; DUAL VII Randomized Clinical Trial

<https://doi.org/10.2337/1817-1134>

Liana K. Billings,^{1,2} Ankur Dash,³
Dilger Gaur,⁴ Alejandra Oviedo,⁵
Helena W. Rodbard,⁶
Nikolaos Tentolaitis,⁷ Randi Gryn,⁸
Natalie Halladin,⁹ and Esteban Jodas⁹



ORIGINAL ARTICLE

OBJECTIVE

In patients with uncontrolled type 2 diabetes on basal insulin, prandial insulin may be initiated. We assessed the efficacy and safety of initiating insulin degludec/liraglutide fixed-ratio combination (IDegLira) versus basal-bolus insulin.

RESEARCH DESIGN AND METHODS

A phase 3b trial examined patients with uncontrolled type 2 diabetes on insulin glargine (IGlar U100) 20–50 units/day and metformin, randomized to IDegLira or IGlar U100 and insulin aspart four or fewer times per day.

RESULTS

Glycated hemoglobin (HbA_{1c}) decreased from 8.2% (66 mmol/mol) to 6.7% (50 mmol/mol) with IDegLira, and from 8.2% (67 mmol/mol) to 6.7% (50 mmol/mol) with basal-bolus (estimated treatment difference [ETD] −0.02% [95% CI −0.16, 0.12]; −0.2 mmol/mol [95% CI −1.7, 1.3]), confirming IDegLira noninferiority versus basal-bolus ($P < 0.0001$). The number of severe or blood glucose–confirmed symptomatic hypoglycemia events was lower with IDegLira versus basal-bolus (risk ratio 0.39 [95% CI 0.29, 0.51]; rate ratio 0.11 [95% CI 0.06, 0.17]). Body weight decreased with IDegLira and increased with basal-bolus (ETD −3.6 kg [95% CI −4.2, −2.9]). Fasting plasma glucose reductions were similar; lunch, dinner, and bedtime self-monitored plasma glucose measurements were significantly lower with basal-bolus. Sixty-six percent of patients on IDegLira versus 67.0% on basal-bolus achieved HbA_{1c} <7.0% (53 mmol/mol). Total daily insulin dose was lower with IDegLira (40 units) than basal-bolus (84 units total; 52 units basal).

CONCLUSIONS

In patients with uncontrolled type 2 diabetes on IGlar U100 and metformin, IDegLira treatment elicited HbA_{1c} reductions comparable to basal-bolus, with statistically superior lower hypoglycemia rates and weight loss versus weight gain.

Diabetes Care Publish Ahead of Print, published online February 26, 2018

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Clinical trial reg. no. NCT02420262, clinicaltrials.gov.

This article contains Supplementary Data on the at <http://care.diabetesjournal.org/lookup/suppl/doi:10.2337/1817-1134/-/DC1>.

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DUAL VII Diseño del estudio

**Sujetos con DMT2
no controlados
con insulina basal
(N=506)**

**IDegLira + metformina
(n=252)**

**IGlar U100 + IAsp (≤4 veces) +
metformina (n=254)**

IDegLira
Dosis inicial: 16 unidades de dosis
Max. dosis: 50 unidades de dosis

IGlar U100
Dosis inicial: dosis pre-estudio
Max. dosis: sin límite

IAsp
Dosis inicial: 4U*
Max. dosis: sin límite



Algoritmo de titulación: IDegLira e IGlar U100

Algoritmo de titulación: IAsp

Crterios de inclusión

- DMT2
- Edad ≥18 años
- HbA_{1c} 7,0–10,0%
- IGlar U100 20–50 U + metformina
- IMC ≤40 kg/m²

GPA media mg/dL	Ajuste de dosis Unidades de dosis o U
<72	−2
72–90	0
>90	+2

GP media (preprandial y al acostarse) mg/dL	Ajuste de dosis U
<72	−1*
72–108	0**
>108	+1***

IDegLira

Bolo-basal

1

pluma¹

2

plumas

1

inyección¹

*2–4

inyecciones³

1

SMPG²

*2–4

SMPG²



Efficacy and Safety of IDegLira Versus Basal-Bolus Insulin Therapy in Patients With Type 2 Diabetes Uncontrolled on Metformin and Basal Insulin; DUAL VII Randomized Clinical Trial

<https://doi.org/10.2337/1817-1134>

Liana K. Billings,^{1,2} Ankur Dash,³ Délier Gault,⁴ Alejandra Oviedo,⁵ Helena W. Rodbard,⁶ Nikolaos Tentolouzis,⁷ Randi Gryn,⁸ Natalie Halladin,⁹ and Esteban Jodanis⁹

OBJECTIVE

In patients with uncontrolled type 2 diabetes on basal insulin, prandial insulin may be initiated. We assessed the efficacy and safety of initiating insulin degludec/liraglutide fixed-ratio combination (IDegLira) versus basal-bolus insulin.

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RESULTS

Glycated hemoglobin (HbA_{1c}) decreased from 8.2% (66 mmol/mol) to 6.7% (50 mmol/mol) with IDegLira, and from 8.2% (67 mmol/mol) to 6.7% (50 mmol/mol) with basal-bolus (estimated treatment difference [ETD] −0.02% [95% CI −0.16, 0.12]; −0.2 mmol/mol [95% CI −1.7, 1.3]), confirming IDegLira noninferiority versus basal-bolus ($P < 0.0001$). The number of severe or blood glucose–confirmed symptomatic hypoglycemia events was lower with IDegLira versus basal-bolus (risk ratio 0.39 [95% CI 0.29, 0.51]; rate ratio 0.11 [95% CI 0.06, 0.17]). Body weight decreased with IDegLira and increased with basal-bolus (ETD −3.57 kg [95% CI −4.2, −2.9]). Fasting plasma glucose reductions were similar; lunch, dinner, and bedtime self-monitored plasma glucose measurements were significantly lower with basal-bolus. Sixty-six percent of patients on IDegLira versus 67.0% on basal-bolus achieved HbA_{1c} <7.0% (53 mmol/mol). Total daily insulin dose was lower with IDegLira (40 units) than basal-bolus (84 units total; 52 units basal).

CONCLUSIONS

In patients with uncontrolled type 2 diabetes on IGlar U100 and metformin, IDegLira treatment elicited HbA_{1c} reductions comparable to basal-bolus, with statistically superior lower hypoglycemia rates and weight loss versus weight gain.

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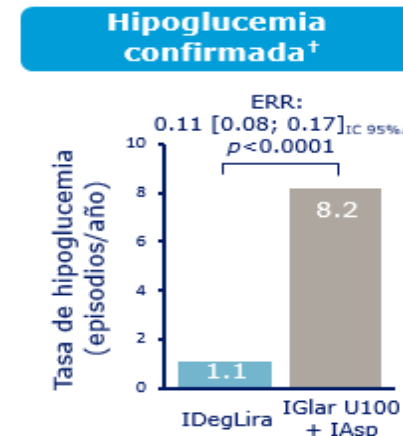
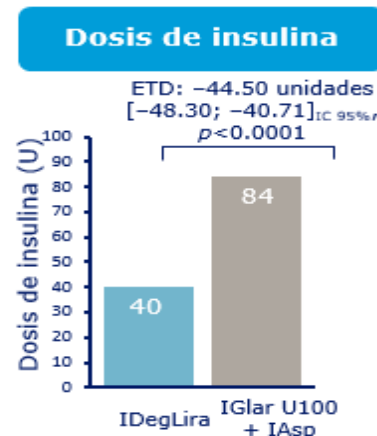
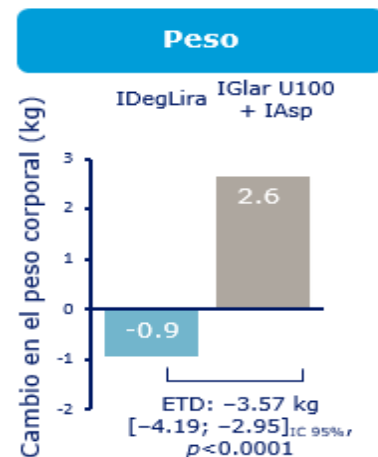
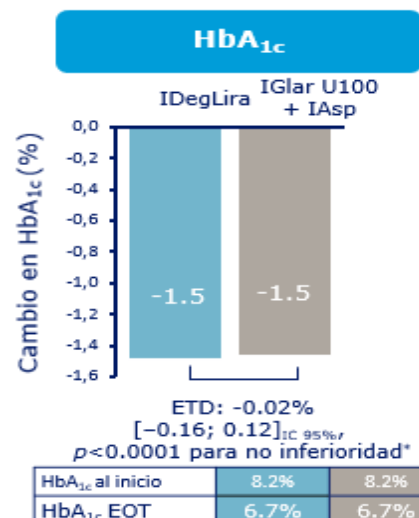
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Clinical trial reg. no. NCT02420262, clinicaltrials.gov.

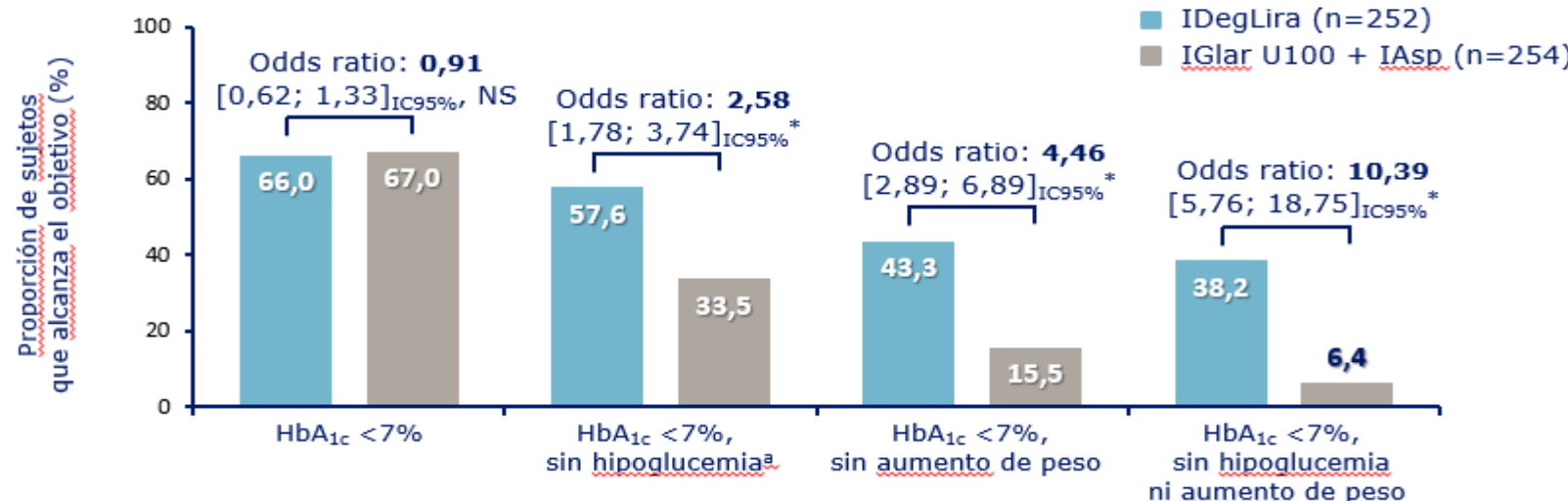
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DUAL VII HALLAZGOS CLÍNICOS CLAVES

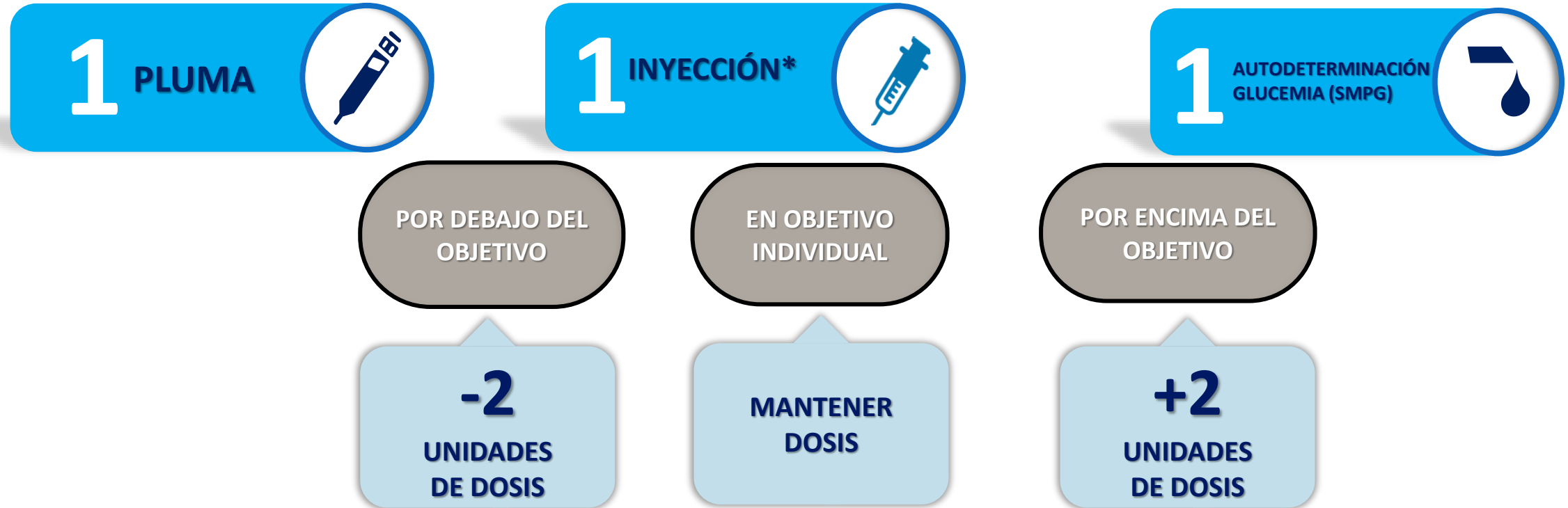


DUAL VII SUBJETOS QUE ALCANZAN LOS OBJETIVOS DE TRATAMIENTO



AJUSTE DE DOSIS DE IDEGLIRA

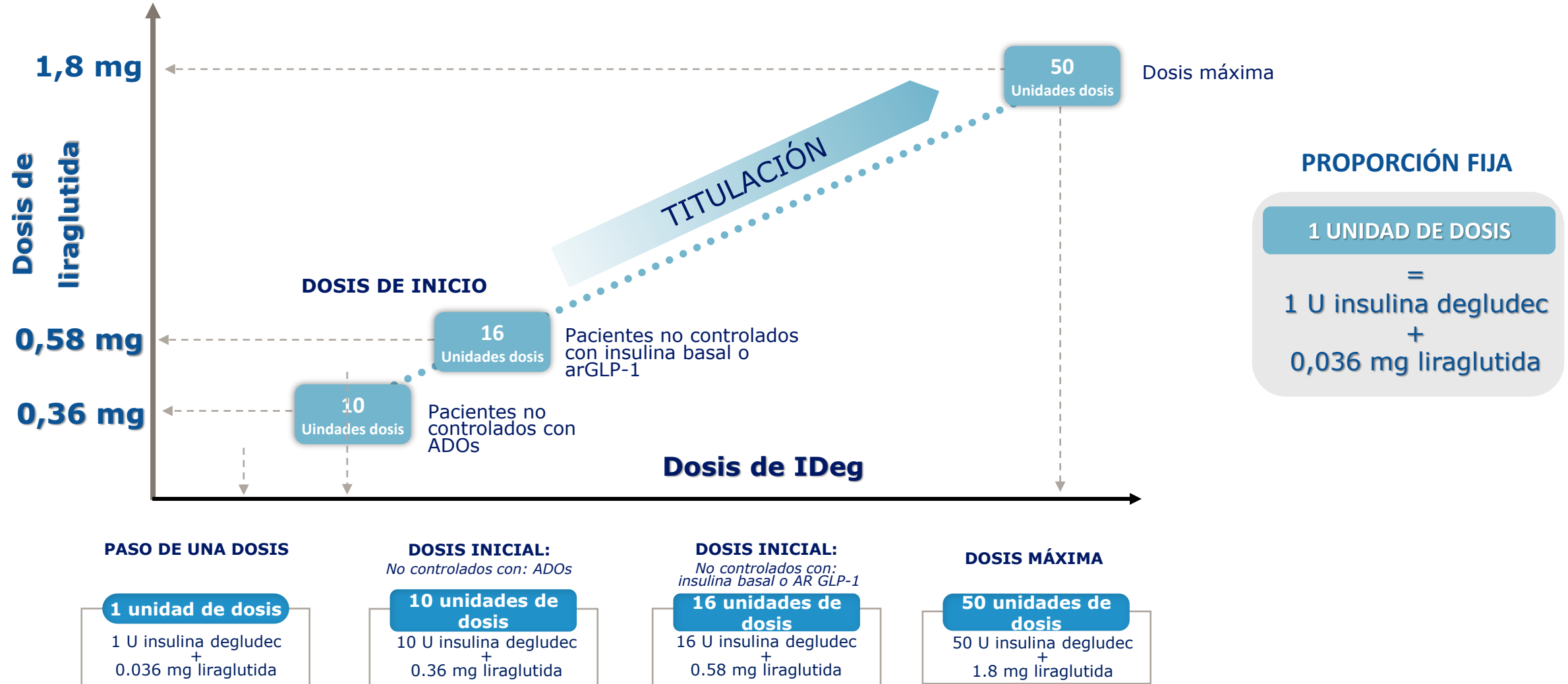
LA DOSIS DE IDEGLIRA SE AJUSTA EN FUNCIÓN DE LAS NECESIDADES DE LOS PACIENTES
TENIENDO EN CUENTA SU GLUCEMIA EN AYUNAS (DESAYUNO)



LAS GUÍAS ADA RECOMIENDAN
UN OBJETIVO DE GLUCEMIA EN AYUNAS Y PREPRANDIAL DE 80 A <130 MG/DL

IDEGLIRA: COMBINACIÓN DIARIA, AJUSTABLE Y EN PROPORCIONES FIJAS

arGLP-1, análogo del receptor del péptido similar a glucagón de tipo 1; IDegLira, insulina degludec/liraglutida; ADO, antidiabético oral
Ficha técnica IDegLira, 2017



Perfil clínico

Nombre: **WAYLON SMITHERS JR.**

Edad: 56 años

Duración de la diabetes: 10 años

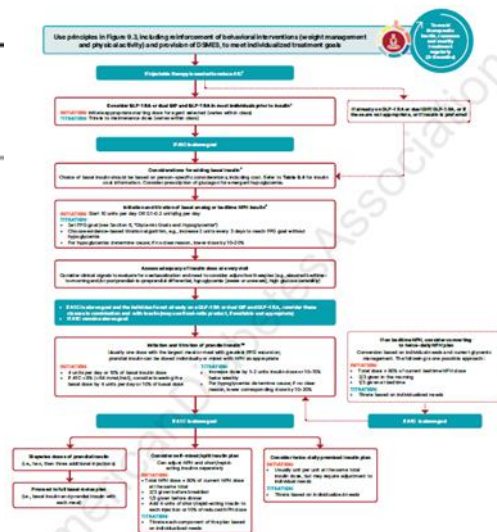
IMC: 23 kg/ m²

Talla 162 CM W: 59,5Kg

Resultados de laboratorio: HbA 1c : 9,2%; Creatinina: 0,8 mg/dL; TFGe: 91/mL/min/1,73 m², RAC 18mg/g c-Ldl 68 mg/dL

Antecedentes médicos previos: Hipertensión, Dislipidemia, hipotiroidismo, depresión actualmente bien controlada con medicación, hipoglucemia no grave poco frecuente (<70 mg/dl) en el año anterior. Pérdida progresiva de peso.

Medicación actual: I.Glar 28 U/día, Perindopril / Amlodipino 5+5mg día, Atorvastatina 40 mg día, levotiroxina 112 mcg día, Duloxetine 30mg día, Metformina 1000mg 2v día, Linagliptina 5 mg día, Dapagliflozina 10mg día.



Diabetes Care.

JANUARY 2025 | VOLUME 48 | SUPPLEMENT 1

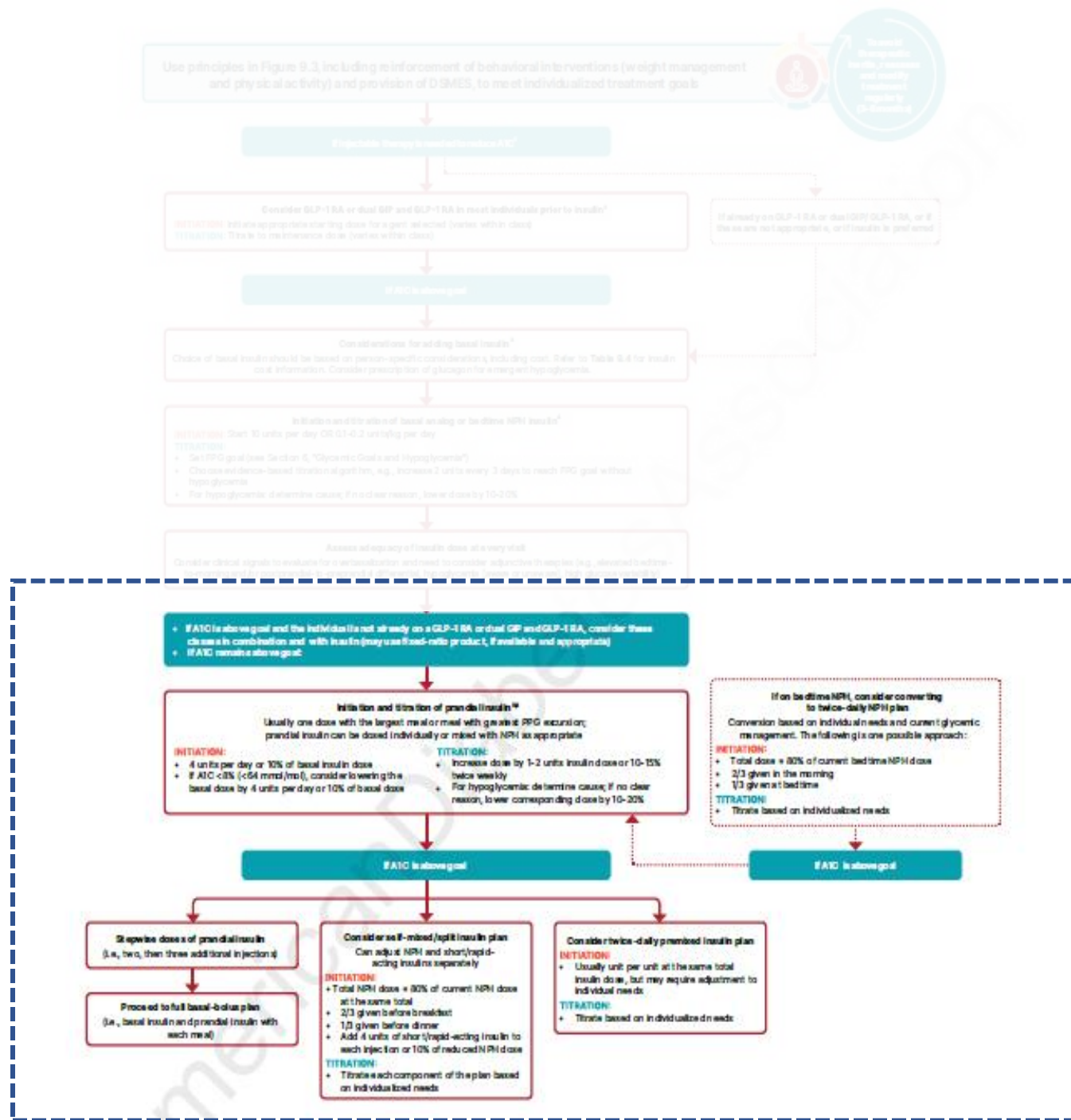
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Standards of Care in Diabetes 2025

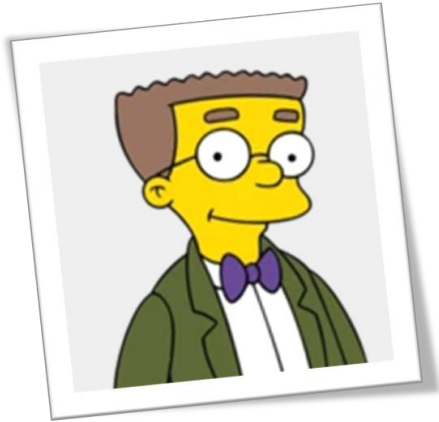
American Diabetes Association.
ISSN 0149-5992

Diabetes Care Volume 48, Supplement 1, January 2025



INTENSIFICACIÓN A TERAPIAS INYECTABLES EN DIABETES TIPO 2.

REFORZAR LAS INTERVENCIONES (CONTROL DE PESO Y ACTIVIDAD FÍSICA) Y PROVISIÓN DE DSMES, PARA CUMPLIR CON LOS OBJETIVOS DE TRATAMIENTO INDIVIDUALIZADOS



Sr. Waylon Smithers Jr.

HbA1c 9,2 %

Categoría de riesgo **MUY ALTO**

IMC: 29 Kg / m²

eTFG E I / V A1

3 ADOs Dosis Máxima +

Insulina basal

- Si está por encima del objetivo de A1C y aún no está en un GLP-1 RA o GIP dual y GLP-1 RA, considere estas clases, ya sea en combinación libre o combinación de proporción fija, con insulina.

AGREGAR INSULINA PRANDIAL

Por lo general, una dosis con la comida más grande o la comida con la mayor excursión de PPG; la insulina puede dosificarse individualmente o mezclarse con NPH según corresponda

INICIACIÓN:

- 4U día o 10% de dosis basal

- Si HbA_{1c} <8% considere disminuir la dosis total en 4U día o 10% de la dosis basal.

TITULACIÓN:

- Aumentar la dosis en 1-2 unidades o 10-15% dos veces por semana

- Para la hipoglucemia determinar causa, si no hay una razón clara DISMINUIR dosis correspondiente en un 10-20%

Si está en NPH a la hora de acostarse, considere convertir al régimen de NPH dos veces al día. Conversión basada en necesidades individuales y actual Control Glicémico.

INICIACIÓN:

Dosis total = 80 % de la dosis actual de NPH

2/3 dado por la mañana

1/3 administrado a la hora de acostarse

TITULACIÓN:

Valorar en función de las necesidades individualizadas

¿TODOS O PASO A PASO?

Si está por encima del objetivo de A1C

Si está por encima del objetivo de A1C

Paso a paso adicional inyecciones de insulina prandial (es decir, dos, luego tres INYECCIONES adicionales)

Continuar con el régimen basal-bolo completo (insulina basal e insulina prandial con cada comida)

Considere el régimen de insulina automezclada/dividida Puede ajustar la NPH y las insulinas de acción corta/rápida

INICIACIÓN:

Dosis total de NPH = 80 % de la dosis actual de NPH

2/3 dado antes del desayuno

1/3 dado antes de la cena

Agregue 4 unidades de insulina de acción corta/rápida a cada inyección o el 10 % de la dosis reducida de NPH

TITULACIÓN:

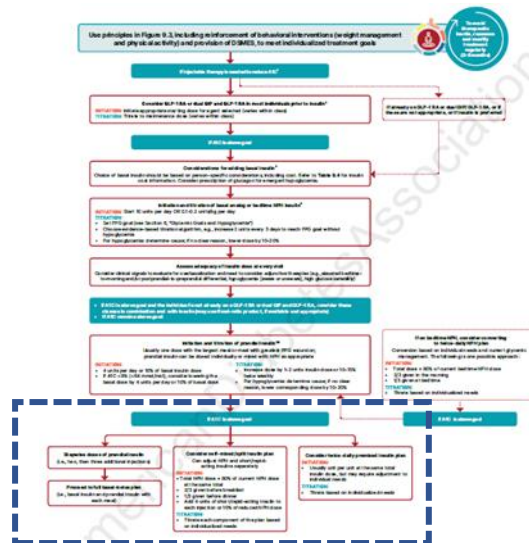
Titular cada componente del régimen en base a las necesidades individualizadas

Considere dos veces al día régimen de insulina premezclada

Usualmente unidad por unidad al mismo total dosis de insulina, pero puede requerir ajuste a necesidades individuales

TITULACIÓN:

Titular en base a necesidades individualizadas





Articles



Treatment intensification with stepwise addition of prandial insulin aspart boluses compared with full basal-bolus therapy (FullSTEP Study): a randomised, treat-to-target clinical trial

Helena W Rodbard, Virginia E Visco, Henning Andersen, Line C Hiort, David H W Shu



Treatment intensification with stepwise addition of prandial insulin aspart boluses compared with full basal-bolus therapy (FullSTEP Study): a randomised, treat-to-target clinical trial

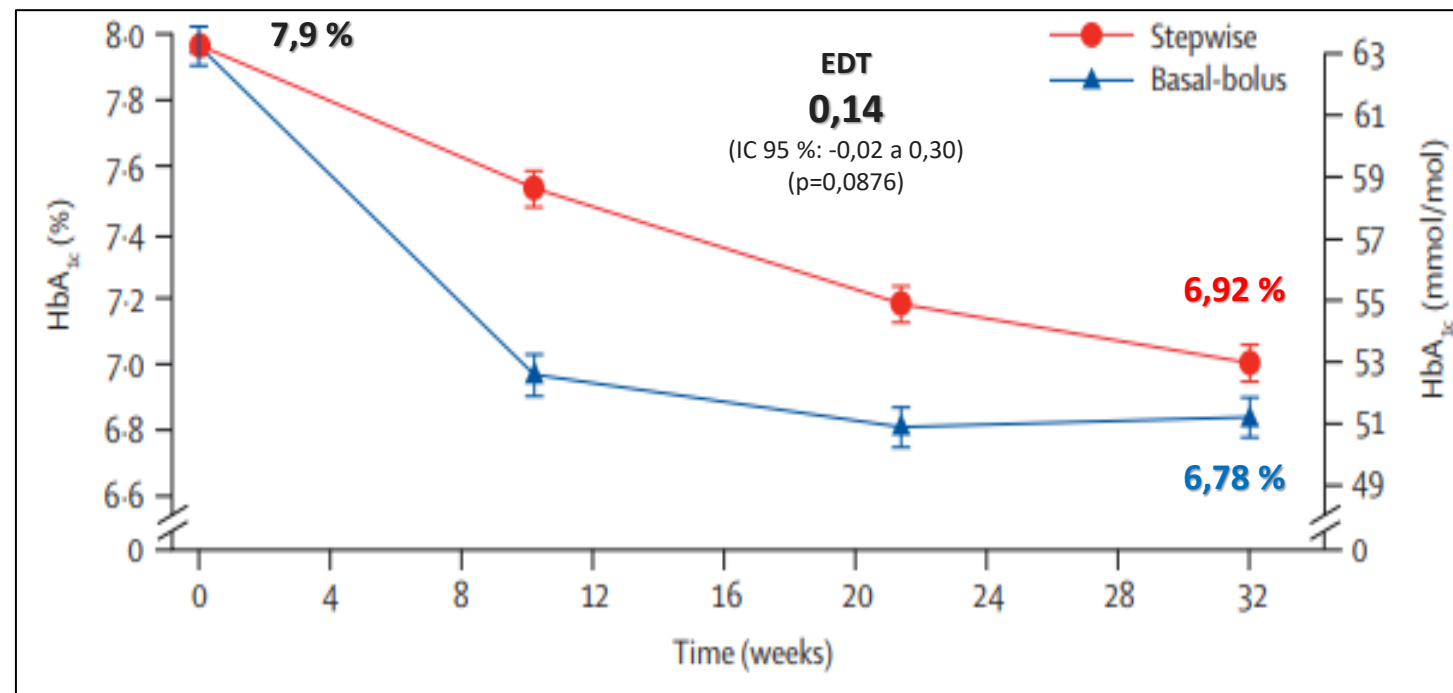
Hilena W Rodbard, Virginia E Vitas, Henning Andersen, Line C Hest, David HW Shi

Summary

Background We compared stepwise addition of bolus insulin with a full basal-bolus regimen in patients with type 2 diabetes inadequately controlled on basal insulin plus oral antidiabetic drugs.

Methods The FullSTEP study was a phase 4, 32-week, randomised, open-label, two-arm, parallel-group, multinational, treat-to-target, non-inferiority trial done at 150 sites across seven countries to assess the effectiveness of a stepwise dosing approach versus a basal-bolus regimen. In this trial, 401 patients (mean age 59.8 years [SD 9.3]; HbA_{1c} 7.9% [63 mmol/mol]; mean diabetes duration 12.6 years [SD 8.0]) were block randomised (ratio 1:1) to receive either stepwise treatment or full basal-bolus treatment. Patients in the basal-bolus group received insulin aspart before every meal throughout the trial. Patients in the stepwise group received one bolus dose with the largest meal, with additional insulin aspart doses before the next largest meal added to their regimen at 11 weeks and 22 weeks if HbA_{1c} remained at 7% or higher. The primary outcome was non-inferiority of stepwise addition of bolus insulin versus complete basal-bolus therapy, as assessed by change in HbA_{1c} from baseline to 32 weeks (non-inferiority margin of 0.4%). This trial is registered with ClinicalTrials.gov, number NCT01165684.

Findings The study was started on Oct 27, 2010, and completed on April 25, 2012. After 32 weeks, HbA_{1c} change from baseline was -0.98% (95% CI -1.09 to -0.87) for the stepwise group and -1.12% (-1.23 to -1.00) for the basal-bolus group; mean treatment difference 0.14 (95% CI -0.02 to 0.30), non-significant ($p=0.0876$). Fewer hypoglycaemic episodes occurred in the stepwise group than in the basal-bolus group (rate ratio 0.58 [95% CI 0.45 to 0.75]; $p<0.0001$). Treatment-emergent adverse events did not differ between the two treatment groups. The most frequently reported treatment-emergent adverse events were nasopharyngitis, influenza, diarrhoea, headache, peripheral oedema, and wrong drug given. Three participants died: two before randomisation and one in the basal-bolus group (due to severe acute myocardial infarction and respiratory tract inflammation).





GLUCOMETRÍAS

4ui



FECHA	AD	DD	AA	DA	AC	DC	3AM
07-ENE	82	150	132	210	190	170	88
08- ENE	102	166	120	250	202	165	120
09- ENE	115	175	156	190	178	228	96
10- ENE	126	180	130	185	192	144	102
11- ENE	95	136	136	330	380	180	116
12- ENE	114	144	150	170	184	168	88
13- ENE	72	40	118	202	165	170	35
14-ENE	82	150	132	120	160	170	88
15- ENE	102	166	120	150	92	165	120
16- ENE	115	175	156	160	168	228	96
17- ENE	126	180	130	145	172	144	102
19- ENE	95	136	136	130	80	180	116
20- ENE	114	144	150	150	174	168	88
21- ENE	72	40	118	92	135	170	35





Management of Hyperglycemia in Type 2 Diabetes, 2018. A Consensus Report by the American Diabetes Association (ADA) and the European Association for the Study of Diabetes (EASD)

Diabetes Care 2018;41:2669–2701 | <https://doi.org/10.2337/dci18-0033>

The American Diabetes Association and the European Association for the Study of Diabetes convened a panel to update the prior position statements, published in 2012 and 2015, on the management of type 2 diabetes in adults. A systematic evaluation of the literature since 2014 informed new recommendations. These include additional focus on lifestyle management and diabetes self-management education and support. For those with obesity, efforts targeting weight loss, including lifestyle, medication, and surgical interventions, are recommended. With regards to medication management, for patients with clinical cardiovascular disease, a sodium–glucose cotransporter 2 (SGLT2) inhibitor or a glucagon-like peptide 1 (GLP-1) receptor agonist with proven cardiovascular benefit is recommended. For patients with chronic kidney disease or clinical heart failure and atherosclerotic cardiovascular disease, an SGLT2 inhibitor with proven benefit is recommended. GLP-1 receptor agonists are generally recommended as the first injectable medication.

The goals of treatment for type 2 diabetes are to prevent or delay complications and maintain quality of life (Fig. 1). This requires control of glycemia and cardiovascular risk factor management, regular follow-up, and, importantly, a patient-centered approach to enhance patient engagement in self-care activities (1). Careful consideration of patient factors and preferences must inform the process of individualizing treatment goals and strategies (2,3).

This consensus report addresses the approaches to management of glycemia in adults with type 2 diabetes, with the goal of reducing complications and maintaining quality of life in the context of comprehensive cardiovascular risk management and patient-centered care. The principles of how this can be achieved are summarized in Fig. 1 and underpin the approach to management and care. These recommendations are not generally applicable to patients with monogenic diabetes, secondary diabetes, or type 1 diabetes, or to children.

Data Sources, Searches, and Study Selection

The writing group accepted the 2012 (4) and 2015 (5) editions of this position statement as a starting point. To identify newer evidence, a search was conducted on PubMed for randomized clinical trials (RCTs), systematic reviews, and meta-analyses

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CONSENSUS REPORT

CONSIDERING ORAL THERAPY IN COMBINATION WITH INJECTABLE THERAPIES



METFORMIN



Continue treatment with metformin

SGLT2i



If on SGLT2i, continue treatment

Consider adding SGLT2i if

- Established CVD
- If HbA_{1c} above target or as weight reduction aid

TZD¹



Stop TZD when commencing insulin OR reduce dose



Beware

- DKA (euglycemic)
- Instruct on sick-day rules
- Do not down-titrate insulin over-aggressively

SULFONYLUREA



If on SU, stop or reduce dose by 50% when basal insulin initiated



Consider stopping SU if prandial insulin initiated or on a premix regimen

DPP-4i



Stop DPP-4i if GLP-1 RA initiated

1. Contraindicated in some countries, consider lower dose. This combination has a high risk of fluid retention and weight gain



CASO 3 : INTENSIFICACIÓN EN PERSONAS NO CONTROLADAS ADECUADAMENTE (HbA 1c > 7%) CON INSULINA BASAL+ BOLO INTENSIVO

Perfil clínico

Nombre: **EDNA KRABAPPEL**

Edad: 58 años

Duración de la diabetes: 14 años

Talla 1,65 m Peso 58 Kg IMC: 21,3 kg/ m²

Resultados de laboratorio: HbA 1c : 7,2%; Creatinina: 0,7 mg/dL;
TFGe: 90,9/mL/min/1,73 m². c-Ldl 52 mg/dL

Antecedentes médicos previos: Hipertensión, Dislipidemia, hipoglucemia frecuentes, Enfermedad Coronaria multivaso # 2 stent DA+CX, Intolerante a la Metformina. Fumadora activa con IPA 20

Medicación actual: I.Glar 42 U/día, Insulina Glulisina 8-8-6, Enalapril 20 mg día, Amlodipino 5mg día, Atorvastatina 80 mg día, levotiroxina 112 mcg día + ASA 100mg día, Bisoprolol 5mg día, Linagliptina 5 mg día, Dapagliflozina 10mg día.

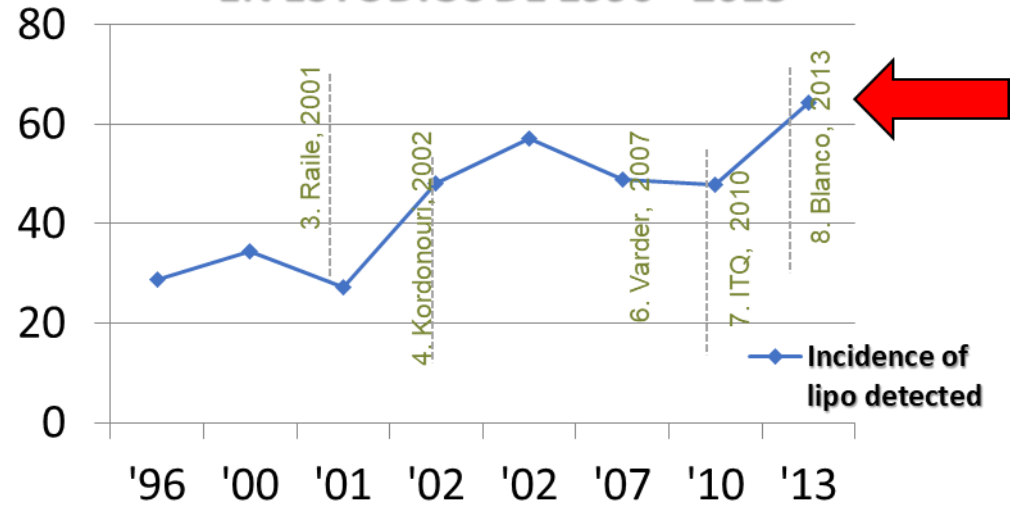


GLUCOMETRÍAS

FECHA	AD	DD	AA	DA	AC	DC	3AM
10 ABR	62	250					
11 ABR			120	250			120
12 ABR					78	228	
13 ABR	156	184					
14 ABR			56	330			
15 ABR					124	224	
16 ABR	72	60					65



INCIDENCIA DE LIPO DETECTADA EN ESTUDIOS DE 1996 - 2013



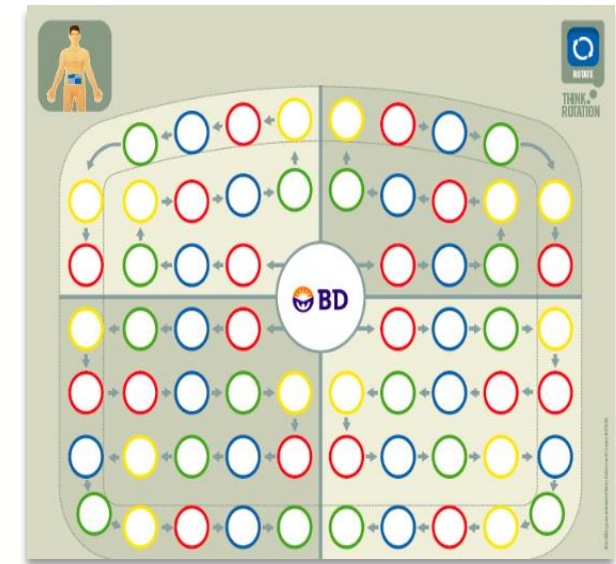
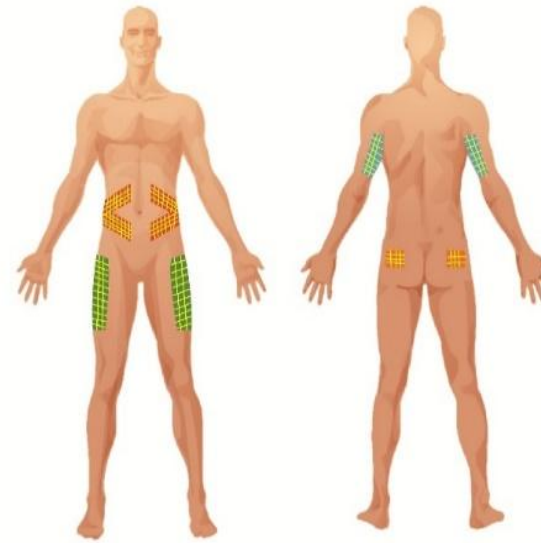
TENER LH E INYECTARSE EN ZONAS CON LH ESTÁ ASOCIADO EN VARIOS ESTUDIOS CON DEFICIENTES RESULTADOS CLÍNICOS, UN MAYOR CONSUMO DE INSULINA Y MAYORES COSTOS AL SISTEMA SANITARIO. 1

VARIABILIDAD GLUCÉMICA

49%

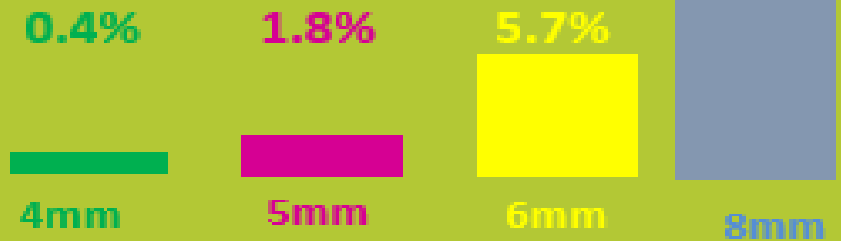
HIPOGLUCEMIAS INEXPLICABLES

39%



PROBABILIDAD CALCULADA DE INYECCIÓN EN EL MUSCULO POR EL LARGO DE LA AGUJA¹⁴

Aguja insertada a 90° sin pliegue en la población adulta



Comparada con Aguja PEN de 4mm

5mm = x5 veces mayor riesgo de inyección IM
6mm = x14 veces mayor riesgo de inyección IM
8mm = x38 veces mayor riesgo de inyección IM



STUDY PROTOCOL

The IDEAL (Insulin therapy DE-intensificAtion with iglarLixi) Randomised Controlled Trial—Study Design and Protocol

Peter Novodvorský · Lenka Thieme · Ivana Laňková · Miloš Mráz · Zoltán J. Taybani ·
Balázs Bótyik · Péter Stella · Miroslav Vytasíl · Felipe Lauand · Mireille Bonnemaiore ·
Martin Haluzík

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ABSTRACT

Introduction: Multiple daily injection insulin regimen (MDI) represents the most intensive insulin regimen used in the management of people with type 2 diabetes (PwT2D). Its efficacy regarding glycaemic control is counterbalanced by the increased risk of hypoglycaemia, frequently observed tendency to weight gain and necessity for frequent glucose monitoring.

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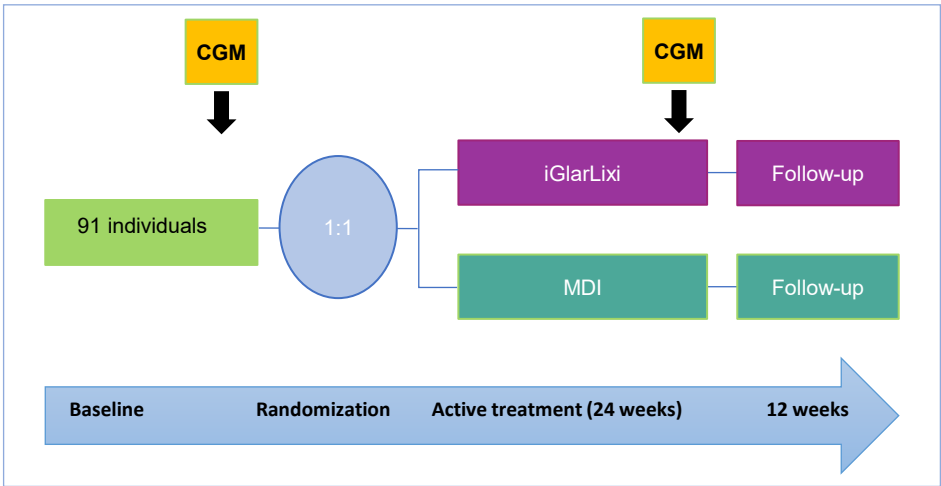
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Recent introduction of novel antidiabetic medications with pleiotropic effects reaching far beyond the reduction of glycaemia (HbA1c), such as the glucagon-like peptide 1 receptor agonist (GLP-1 RA), has significantly widened the therapeutic options available for management of T2D. Consequently, there is currently a substantial number of PwT2D for whom the MDI regimen was initiated at a time when no other options were available. Yet, in present times, these individuals could benefit from simplified insulin regimens ideally taking advantage of the beneficial effects of the novel classes of antidiabetic medications. iGlarLixi (Suliqua®) is a once-daily fixed-ratio combination of basal insulin analogue glargine 100 U/ml and a GLP-1 RA lixisenatide.

Methods: Insulin therapy DE-intensificAtion with iglarLixi (IDEAL) is a six-centre, open-label, parallel-group, active comparator, phase IV randomised controlled trial with a 24-week active treatment period examining the efficacy and safety of MDI regimen de-intensification with once-daily administration of iGlarLixi versus MDI regimen continuation in PwT2D on a background therapy with metformin ± sodium-glucose cotransporter 2 inhibitor.

Planned Outcomes: The primary objective is to compare the effects of MDI therapy de-intensification with iGlarLixi versus MDI regimen continuation regarding glycaemic control (HbA1c). Secondary objectives include detailed evaluation



Previous daily basal insulin dose*		≥ 10 to <20 units/day	≥ 20 to <30 units/day	≥30 to ≤50 units/day
Initial iGlarLixi dose	iGlarLixi 100/50 (pen A)	≥ 10 to <20 dose steps/day	20 dose steps/day	
	iGlarLixi 100/33 (pen B)			30 dose steps/day

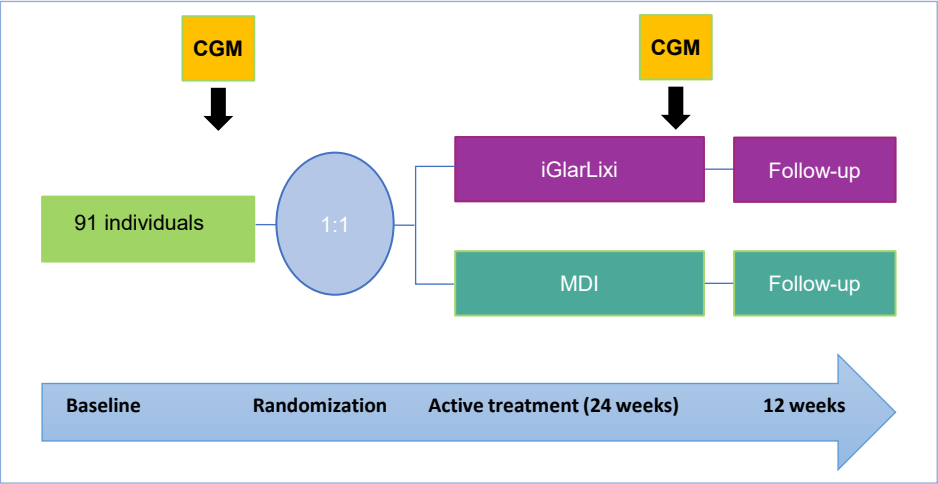
	Total N = 91	iGlarLixi N = 45	MDI N = 46
Age (years), Mean (SD)	66 (9)	65 (9)	67 (9)
Female, n (%)	20 (22%)	11 (24%)	9 (20%)
Race, n (%)			
White	87 (96%)	43 (96%)	44 (96%)
Black	0 (0%)	0 (0%)	0 (0%)
Other	4 (4.4%)	2 (4.4%)	2 (4.3%)
Body weight (kg), Mean (SD)	101 (17)	100 (18)	102 (17)
BMI (kg/m ²), Mean (SD)	33.5 (5.5)	33.7 (5.5)	33.4 (5.5)
Duration of diabetes (years), Mean (SD)	17 (9)	17 (9)	18 (9)
HbA1c, Mean (SD)	7.88 (1.01)	7.96 (1.00)	7.80 (1.01)
Basal insulin, n (%)			
Degludec	18 (20%)	10 (22%)	8 (17%)
Detemir	17 (19%)	12 (27%)	5 (11%)
iGlar U100	23 (25%)	8 (18%)	15 (33%)
iGlar U300	32 (35%)	14 (31%)	18 (39%)
NPH insulin	1 (1.1%)	1 (2.2%)	0 (0%)
Prandial insulin, n (%)			
Prandial analog	83 (91%)	41 (91%)	42 (91%)
Human prandial insulin	8 (8.8%)	4 (8.9%)	4 (8.7%)
Total daily dose of insulin (IU), Mean (SD)	61 (21)	63 (20)	59 (22)
Oral antidiabetic agents, n (%)			
Metformin	72 (79%)	36 (80%)	36 (78%)
SGLT2 inhibitors	41 (45%)	20 (44%)	21 (46%)
Glitazones	2 (2.2%)	1 (2.2%)	1 (2.2%)
Sulphonylurea derivatives	0 (0%)	0 (0%)	0 (0%)
DPP4 inhibitors	0 (0%)	0 (0%)	0 (0%)



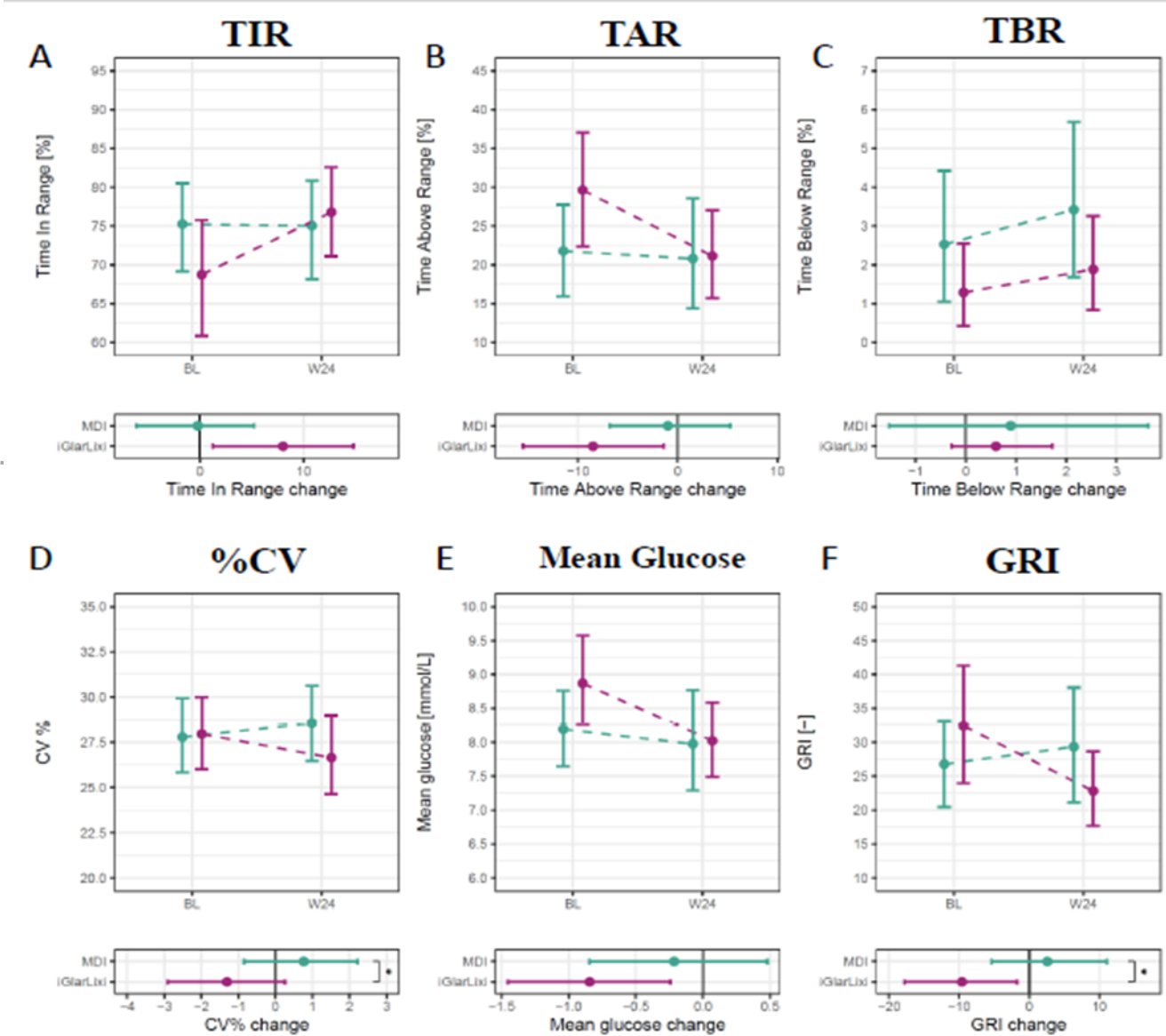
STUDY PROTOCOL

The IDEAL (Insulin therapy DE-intensification with iglarLixi) Randomised Controlled Trial—Study Design and Protocol

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Previous daily basal insulin dose*		≥ 10 to <20 units/day	≥ 20 to <30 units/day	≥30 to ≤50 units/day
Initial iGlarLixi dose	iGlarLixi 100/50 (pen A)	≥ 10 to <20 dose steps/day	20 dose steps/day	
	iGlarLixi 100/33 (pen B)			30 dose steps/day



UNA PROPUESTA PARA LA SIMPLIFICACIÓN DE REGÍMENES COMPLEJOS DE INSULINA CON IDEGLIRA

Diabetes Ther (2023) 14:1959–1976
https://doi.org/10.1007/s13300-023-01471-9



PRACTICAL APPROACH

Simplification of Complex Insulin Regimens with IdegLira in People with Type 2 Diabetes: Literature Review and Clinical Recommendations

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ABSTRACT

Introduction: This study developed a simple algorithm based on clinical results described in medical literature and which allows one to simplify complex insulin regimens with IdegLira to avoid adverse events related to the complexity of some insulin treatments.

Methods: We conducted a systematic review of the literature that allowed us to identify studies that evaluated the clinical result of simplifying complex insulin regimens. The authors reviewed

the common factors these simpler regimens had, including the type of patients who used them.

Results: We found nine clinical studies published between 2017 and 2022, eight performed in Europe and one in Latin America. The monitoring time of the studies ranged between 3 and 18 months. The size of the study populations was between 61 and 611 patients (the latter was in five countries). In all studies, HbA1c decreased by 0.6–1.7% and the weight decreased by 0.1–3.11 kg.

Conclusions: On the basis of the findings of these studies, we made some recommendations for clinical practice to simplify treatment. The results of these studies support an algorithm that simplifies the treatment of complex insulin regimens.

Keywords: Basal insulin; Beta cell function; GLP-1 analogue; Insulin therapy; Liraglutide; Systematic review

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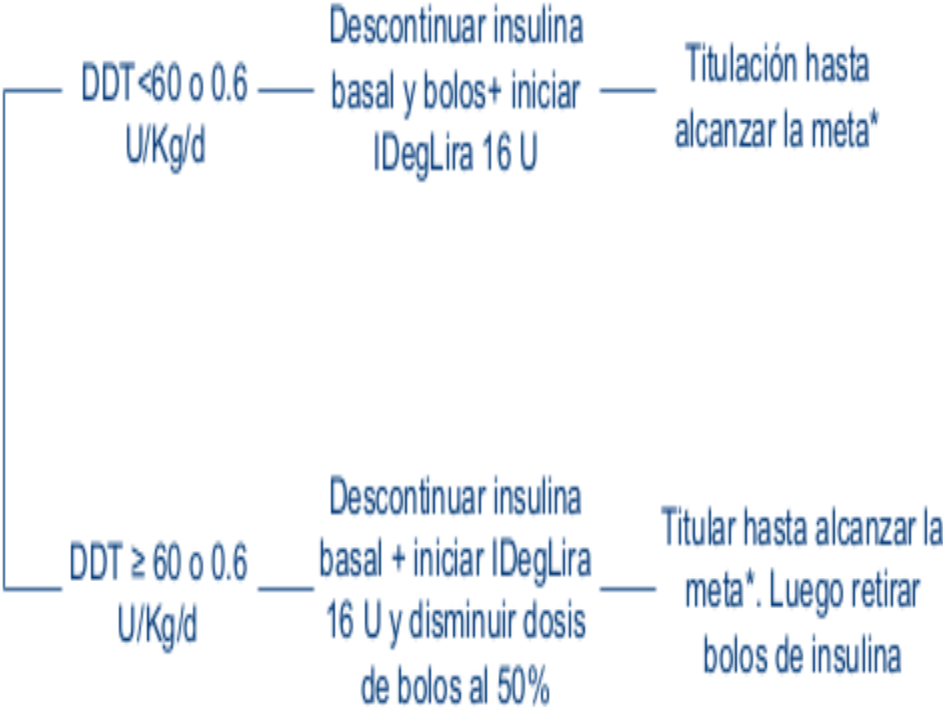
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Pacientes con DM2 en terapia con
múltiples dosis de insulina y ausencia
de disfunción de célula beta



*Descontinuar inhibidores DPP-4, insulina basal y otros ar-GLP1 cuando se inicia IdegLira.

Titular con base a la glucosa plasmática en ayunas:

- De acuerdo con ADA (American Diabetes Association): 80-130 mg/dl
- De acuerdo con ensayos DUAL: 72-90 mg/dl
- En adultos mayores frágiles: 110-150 mg/dl

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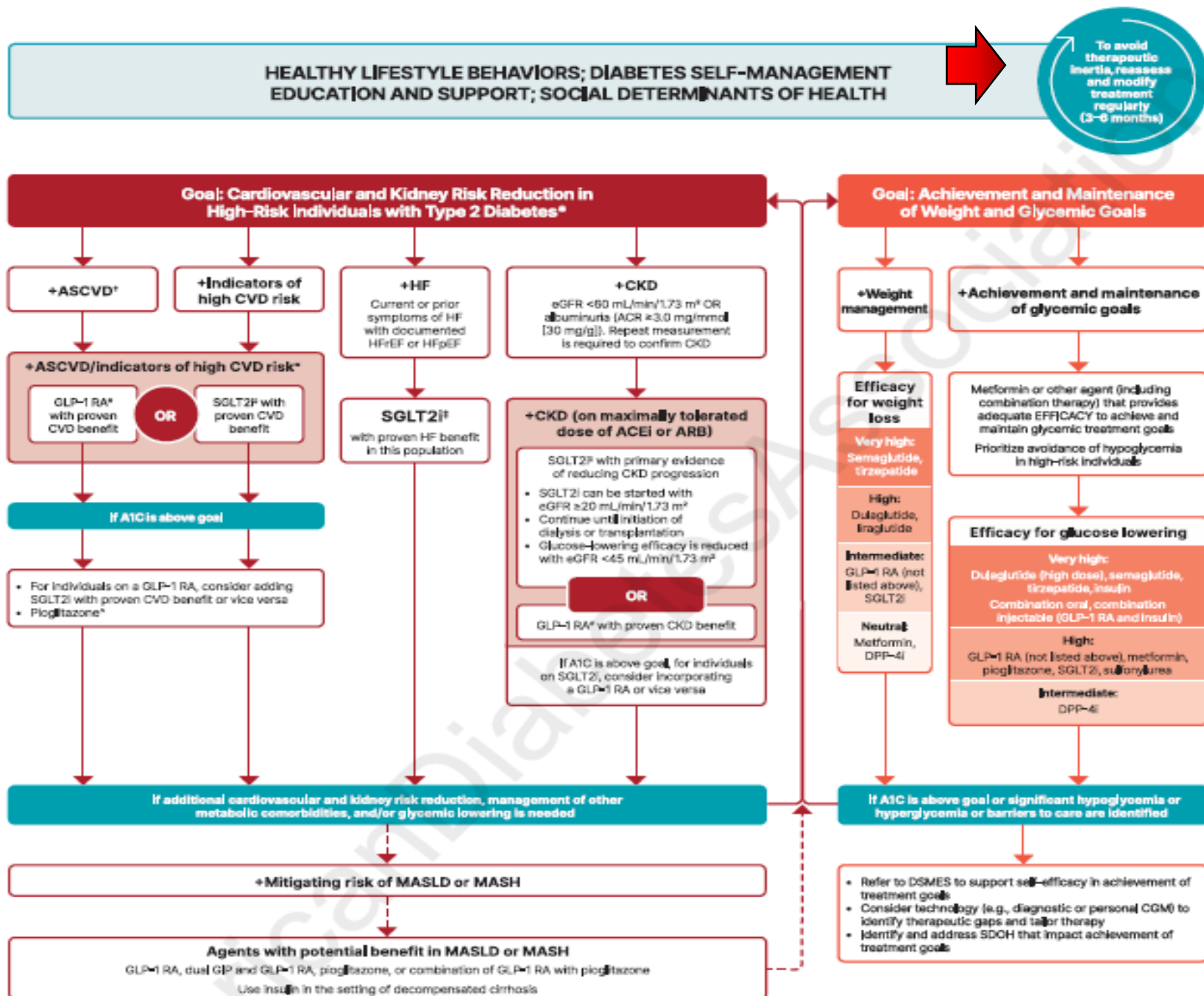


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Use of Glucose-Lowering Medications in the Management of Type 2 Diabetes



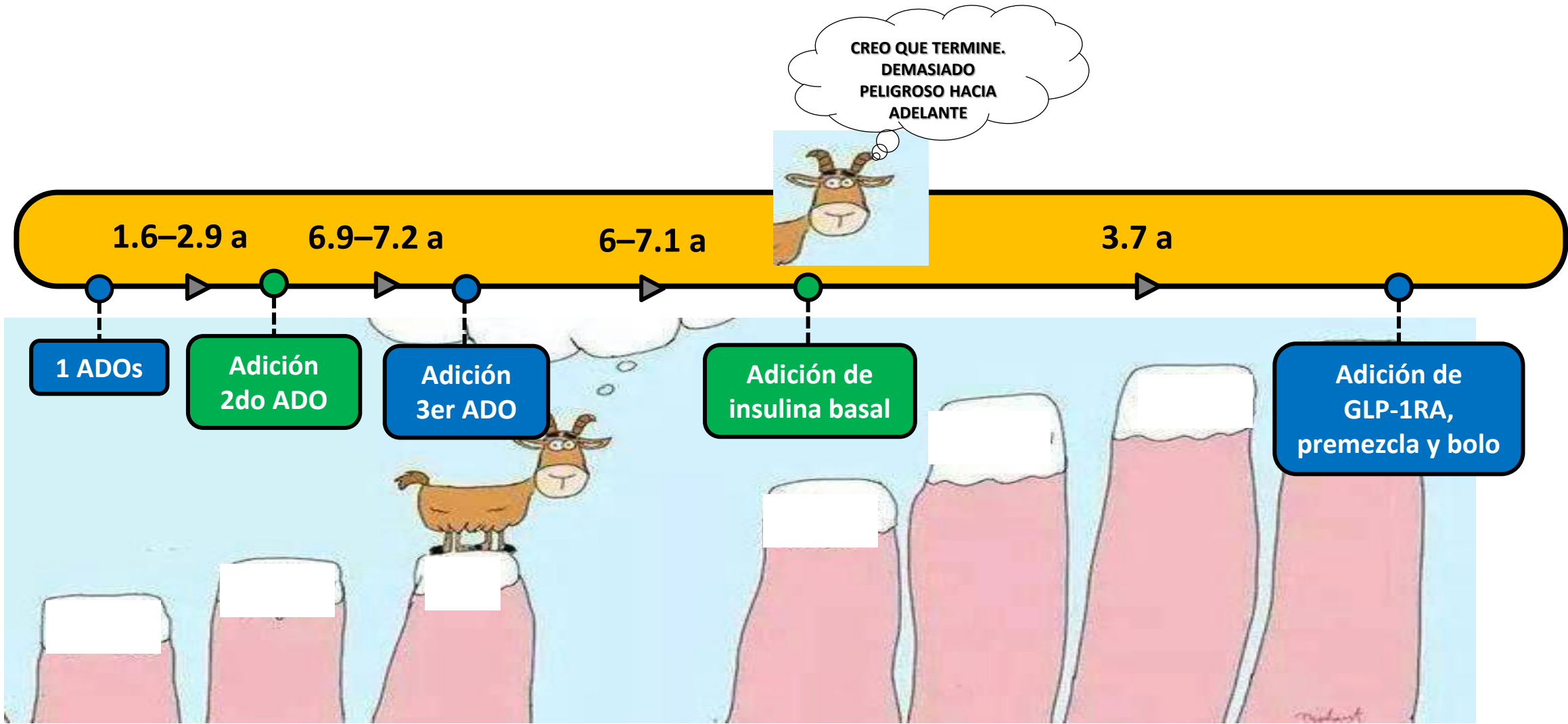
Clinical inertia to insulin initiation and intensification in the UK: A focused literature review

Kamlesh Khunti^{a,*}, David Millar-Jones^b



Clinical inertia with regard to intensifying therapy in people with type 2 diabetes treated with basal insulin

K. Khunti^{1,†}, A. Nikolajsen², B. L. Thorsted², M. Andersen³, M. J. Davies¹ & S. K. Paul^{4,†}



H O M E R A T R I X





That's all Folks!